

An Expanded Implementation-Ready Mathematical Companion To Fixed Sparse-Grid Quadrature Filtering And Its Analytical Gradient

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Abstract

This note develops fixed sparse-grid quadrature filtering as an implementation-ready high-dimensional filtering proposal for the regime where exact nonlinear filtering is unavailable but a deterministic Gaussian-surrogate value-and-gradient target is still scientifically useful. Starting from the sparse-grid quadrature nonlinear filter of Jia, Xin, and Cheng [1], the note reconstructs the Gaussian approximation and sparse-grid formulas in source order, then defines a fixed approximate likelihood scalar together with an analytical derivative of that same scalar on a declared branch. The document is written for both review and implementation: it states the exact filtering target, the narrower carried Gaussian surrogate, the fixed cloud-construction rule, the per-step value recursion, the same-scalar branch contract, the analytical gradient recursion, and the finite-difference validity rule. The central limitation is explicit. The reported scalar is not the exact nonlinear filtering likelihood, and the method does not preserve general non-Gaussian posterior shape; it is a deterministic Gaussian-surrogate lane whose usefulness depends on whether one carried Gaussian and a fixed sparse-grid moment engine are adequate for the intended regime.

1 Reader Orientation, Scope, And How To Use This Note

This note is written for three readers at once:

1. implementation engineers who need an auditable mathematical contract,
2. mathematically trained first-year-PhD-level workers who must implement the algorithm correctly from the note, and
3. general scientific reviewers who need to judge whether the proposal is coherent, bounded, and worthy of approval.

The document therefore serves two roles at once. It is a mathematical companion note for one deterministic approximate filtering lane, and it is an implementation handoff document for that lane. It is not a general survey of nonlinear filtering and it is not a full software engineering manual.

Background assumed. The reader is assumed to know multivariate Gaussian notation, basic state-space-model notation, Jacobian-style derivative notation, and the use of a Cholesky factor for a positive-definite covariance matrix. The note does not assume prior familiarity with fixed-branch derivative contracts, sparse-grid cloud merging, or same-scalar finite-difference validation; those are defined here.

What this note provides internally. The note defines:

1. the exact filtering target and the narrower approximate object carried by FixedSGQF,
2. the fixed sparse-grid cloud construction in standardized coordinates,
3. the one-step value recursion and the accumulated deterministic approximate likelihood scalar,
4. the fixed-branch analytical derivative of that same scalar,
5. the same-scalar finite-difference validity rule, and
6. worked numerical checks that an implementation can use as direct oracles.

How to read the note. Sections 2–3 state the problem, the selected approximation lane, and the exact-versus-approximate object map. Sections 10, 11, and 12 define the fixed cloud and the value recursion. Sections 14 and 15 define the saved branch and the analytical gradient. Section 19 states the finite-difference audit. Implementation workers should treat the equations, contracts, boxed algorithms, and worked example as the authoritative coding material.

What this note does not claim. The note does not claim exact nonlinear filtering, exact nonlinear likelihood evaluation, or faithful preservation of arbitrary posterior multimodality, skewness, or hard all-coordinate interaction structure. It defines one deterministic Gaussian-surrogate lane and makes its approximation boundary explicit.

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2 The High-Dimensional Filtering Problem And The FixedSGQF Lane

The scientific problem is high-dimensional nonlinear filtering. In a general state-space model, the exact Bayesian filter propagates a full probability law through time. Once the transition or observation map is nonlinear, that law is usually no longer describable by a finite list of moments, and in high dimension it becomes unrealistic to treat exact function-valued propagation as a practical computational object. A review panel should therefore not read this report as an attempt to rescue exactness by notation. The question is narrower and more honest: what deterministic approximation lane is mathematically coherent, transparent about its compromises, and still useful enough to deserve serious consideration?

The lane proposed here is FixedSGQF, short for fixed sparse-grid quadrature filtering. It is motivated by a common tension in scientific filtering work. On one side, full non-Gaussian methods can represent richer posterior geometry, but they often carry heavier approximation objects and more complicated smoothness or implementation burdens. On the other side, one may still want a deterministic approximate likelihood surrogate whose value is reproducible, whose dependence on model parameters can be differentiated analytically on a declared branch, and whose numerical behavior is inspectable without Monte Carlo noise. FixedSGQF belongs to the second lane.

The proposal is not to store the full filtering density. It is to store one Gaussian surrogate at each time, to evaluate the nonlinear moments required by that Gaussian projection with a deterministic sparse-grid rule, and then to freeze the grid, merge, and covariance-factor branches so that the reported gradient differentiates the same scalar that the reported value routine evaluates. The gain from that choice is transparency: every approximation stage can be named, inspected, and tested directly. The cost is equally explicit: a single Gaussian carried object cannot preserve general multimodal or strongly skewed posterior structure, even when the quadrature itself is accurate for the moments it is asked to approximate.

This is why FixedSGQF should not be read as a weaker version of the Zhao–Cui squared tensor-train construction. The two proposals serve different goals. FixedSGQF is the deterministic Gaussian-surrogate lane with an analytical same-scalar gradient. Zhao–Cui squared TT is the richer non-Gaussian density-approximation lane. A serious high-dimensional filtering program may want both. The purpose of this report is to show that the FixedSGQF lane is self-contained, mathematically explicit, and implementable enough to stand on its own merits.

For the review chair, the decision test is therefore the following. This lane is worth advancing if the program needs a deterministic, inspectable, and analytically differentiable likelihood surrogate whose approximation boundary is fully exposed, even while richer non-Gaussian density lanes remain necessary for problems where one carried Gaussian is too weak. The report must make that trade-off explicit enough that approval does not depend on optimism or tacit background knowledge.

3 FixedSGQF In One Page

FixedSGQF is built from three deliberate approximation decisions.

First approximation: Gaussian carried state. At time t , the exact filter carries the full conditional law $p_\theta(x_t \mid y_{1:t})$. FixedSGQF instead carries only one Gaussian, recorded by a mean and

covariance pair (m_t, P_t) . This is the main structural narrowing of the proposal. The report must therefore keep separate throughout what belongs to the exact Bayesian filter and what belongs only to its Gaussian surrogate.

Second approximation: sparse-grid moment evaluation. Once the carried object has been narrowed to a Gaussian, the filter still needs nonlinear prediction and observation moments. FixedSGQF evaluates those moments with a deterministic sparse-grid quadrature rule built in standard Gaussian coordinates and then mapped into physical state coordinates by the current covariance factor. The rule is fixed before likelihood evaluation, so the moments are deterministic functions of the declared branch and the model parameters.

Third approximation: fixed-branch differentiation. The scalar differentiated in this note is not the derivative of an adaptive grid selection algorithm. It is the derivative of a declared deterministic scalar computed on a frozen branch. Grid level, merged cloud, duplicate-node policy, covariance-factor family, and veto rules are fixed first. Within that branch, the numerical values of the moments, gains, and likelihood contributions still depend on θ , but the structural route through which they are computed does not change.

The resulting scalar is

$$\hat{\ell}_T^{\text{FSGQ}}(\theta) = \sum_{t=1}^T \hat{\ell}_t^{\text{FSGQ}}(\theta), \quad (3.1)$$

where each increment is the Gaussian innovation log likelihood implied by the sparse-grid moment approximation. The exact object and the approximate object are therefore different in kind:

$$\begin{aligned} \text{exact object:} & \quad p_\theta(x_t \mid y_{1:t}), \\ \text{approximate carried object:} & \quad \mathcal{N}(m_t, P_t), \\ \text{accumulated deterministic scalar:} & \quad \hat{\ell}_T^{\text{FSGQ}}(\theta). \end{aligned} \quad (3.2)$$

That distinction is not bookkeeping. It is the central scientific claim of the report: FixedSGQF is worthwhile only if one judges that this narrower, deterministic, Gaussian-projection lane is still scientifically useful in the regime under study.

Approximation ladder. The rest of the report unfolds four commitments, each narrower than the one before it:

1. **Exact target:** the scientific target is the exact nonlinear filtering law $p_\theta(x_t \mid y_{1:t})$.
2. **Gaussian carried surrogate:** FixedSGQF carries only one Gaussian summary $\mathcal{N}(m_t, P_t)$ instead of the full law.
3. **Sparse-grid moment closure:** the moments defining that Gaussian surrogate are approximated by a deterministic sparse-grid rule.
4. **Fixed-scalar derivative contract:** the reported gradient is the derivative of the declared deterministic surrogate scalar on a frozen branch, not the derivative of a live adaptive algorithm.

A chair should be able to retain this ladder after one reading: exact target, Gaussian surrogate, sparse-grid moment closure, fixed-scalar derivative contract. Every later section exists to unpack one rung of this scheme.

Operational summary for implementers. A first implementation should preserve the following runtime order.

1. Build and save one fixed sparse-grid cloud in standardized Gaussian coordinates.
2. Start from an initial Gaussian surrogate (m_0, P_0) .
3. At each time step, place the previous cloud with the current factor, push points through the transition, and compute predictive moments.
4. If the predictive covariance fails the declared branch, stop and report a predictive-branch failure.
5. Place the predictive cloud, push points through the observation map, and compute $\bar{z}_t$, S_t , and $C_{xz,t}$.
6. If the innovation covariance fails the declared branch, stop and report an innovation-branch failure.
7. Form the one-step Gaussian innovation log likelihood, update the running scalar, and update the carried Gaussian.
8. If gradients are requested, propagate the same fixed-branch derivative recursion on the same cloud and under the same branch policies.

This summary is not a replacement for the later derivation. It is the short execution picture a junior implementer should retain before reading the longer sections.

4 Approximation Hierarchy And Coordinate Walk

The method in this note is easiest to read if those four commitments are kept in the same order in which a program would apply them. Start with the exact Bayesian filter. At time t , the exact predictive density is a function

$$x_t \mapsto p_\theta(x_t \mid y_{1:t-1}),$$

and the exact filtering density after seeing y_t is

$$x_t \mapsto p_\theta(x_t \mid y_{1:t}).$$

Those are full functions on $\mathbb{R}^{n_x}$. A general nonlinear observation can make them skewed, curved, or multimodal. FixedSGQF does not store those functions. It replaces each carried density by one Gaussian:

$$p_\theta(x_t \mid y_{1:t}) \text{ is represented only by } (m_t, P_t).$$

The first approximation is therefore a Gaussian projection. The second approximation is quadrature: the Gaussian moments needed for the projection are computed by a deterministic sparse-grid sum. The third approximation is the fixed-branch contract: the sparse-grid cloud, duplicate merge, covariance factor branch, and failure exits are held fixed so that the analytical derivative differentiates the same scalar that the value routine evaluates.

The coordinates are:

$$\xi \in \mathbb{R}^{n_x}, \quad \xi \sim \mathcal{N}(0, I_{n_x}) \quad \text{standard quadrature coordinate,} \quad (4.1)$$

$$x = m + C\xi, \quad CC^\top = P \quad \text{physical state coordinate under a carried Gaussian,} \quad (4.2)$$

$$a = f_\theta(x), \quad \text{transition image used for prediction,} \quad (4.3)$$

$$z = h_\theta(x), \quad \text{observation image used for the innovation.} \quad (4.4)$$

The sparse grid is built once in ξ -space. The same standardized nodes $\xi^{(r)}$ are then placed into the physical state space by the current Gaussian factor C . This is why the same saved cloud can be reused at every time while the physical points move as m_t , P_t , and θ move.

The likelihood scalar is also an approximation hierarchy. The exact predictive likelihood factor is

$$p_\theta(y_t \mid y_{1:t-1}) = \int g_\theta(y_t \mid x_t) p_\theta(x_t \mid y_{1:t-1}) dx_t. \quad (4.5)$$

FixedSGQF instead forms a Gaussian innovation model

$$Y_t \mid y_{1:t-1} \approx \mathcal{N}(\bar{z}_t, S_t), \quad (4.6)$$

where $\bar{z}_t$ and S_t are sparse-grid approximations of the moments of $h_\theta(X_t^-) + \varepsilon_t$. The scalar differentiated later is

$$\hat{\ell}_T^{\text{FSGQ}}(\theta) = \sum_{t=1}^T \log \mathcal{N}(y_t; \bar{z}_t(\theta), S_t(\theta)). \quad (4.7)$$

Equation (4.7) is useful because it is deterministic, reproducible, and differentiable on a fixed branch. It is narrower than the exact likelihood in (4.5). This distinction is not a weakness in the derivation; it is the contract that makes the method implementable and auditable.

Why moderate sparse-grid level can be plausible. Suppose the observation map is not an arbitrary fully coupled function of all state coordinates, but has a low effective interaction order. A typical block-local observation has the form

$$h(x) = \sum_{j=1}^B h_j(x_{\mathcal{I}_j}) + \sum_{(j,k) \in \mathcal{E}} h_{jk}(x_{\mathcal{I}_j}, x_{\mathcal{I}_k}), \quad (4.8)$$

where each block $\mathcal{I}_j$ is small and the edge set $\mathcal{E}$ is sparse. The Gaussian moments of (4.8) depend strongly on low-order coordinate interactions and weakly on most high-order interactions. A sparse-grid rule of modest level is designed to spend points on lower-order tensor interactions before it spends points on the full tensor product. That is the mathematical reason the point count can be plausible. It is not a theorem that the posterior is easy. If the likelihood has important all-coordinate interactions, narrow ridges, or separated modes, a fixed moderate sparse grid can fail even though its point count is attractive.

A physical analogy without changing the mathematics. In a chemical kinetics model, a measured spectrum or reaction rate often depends strongly on a few combinations of concentrations and rate constants, and only weakly on many nuisance directions. A deterministic sparse-grid rule can be plausible in such a setting because it samples low-order coordinate interactions systematically. The filter still checks the result by innovation diagnostics, positive-definiteness, level ladders, and finite differences. The argument is "this interaction structure makes a moderate grid worth trying," not "chemistry makes the approximation exact."

5 What This Note Computes

This section is the report’s first exact scope statement. It answers three questions that must remain distinct throughout the rest of the document: what object is carried from time step to time step, what scalar is accumulated, and what information about the full posterior law is deliberately discarded.

The object computed here is deliberately narrow. At time t , before observing y_t , the filter carries one Gaussian approximation

$$X_{t-1} \mid y_{1:t-1} \approx \mathcal{N}(m_{t-1}, P_{t-1}). \quad (5.1)$$

It then uses a deterministic sparse-grid quadrature rule to approximate the Gaussian moments needed by a Kalman-style projection update. The output after observing y_t is again only

$$X_t \mid y_{1:t} \approx \mathcal{N}(m_t, P_t), \quad (5.2)$$

together with the approximate Gaussian innovation log likelihood

$$\widehat{\ell}_t = -\frac{1}{2} \left\{ \log \det S_t + v_t^\top S_t^{-1} v_t + n_y \log(2\pi) \right\}. \quad (5.3)$$

The fixed-SGQF proposal is to use the sum

$$\widehat{\ell}_T^{\text{FSGQ}}(\theta) = \sum_{t=1}^T \widehat{\ell}_t^{\text{FSGQ}}(\theta) \quad (5.4)$$

as a deterministic approximate likelihood scalar and to differentiate that same scalar analytically.

This note therefore computes an approximate likelihood and a Gaussian projection recursion. It does not compute the full nonlinear posterior law. Sparse-grid exactness is exactness of a quadrature formula for selected Gaussian-weighted polynomial integrands. It is not a guarantee that the posterior is Gaussian, unimodal, or accurately represented by (m_t, P_t) .

A scalar cell that keeps us honest. Let

$$X \sim \mathcal{N}(0, P), \quad Y = X^2 + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, R). \quad (5.5)$$

For $y > 0$ and small R , the exact posterior density is proportional to

$$\exp \left[-\frac{x^2}{2P} - \frac{(y - x^2)^2}{2R} \right], \quad (5.6)$$

which can have two separated lobes near $+\sqrt{y}$ and $-\sqrt{y}$. A Gaussian projection filter can compute accurate moments of X^2 , but after the update it still stores only one Gaussian. This is not a bug in the quadrature rule. It is the modeling choice in (5.2).

The same limitation explains the role of fixed SGQF in the high-dimensional program. It is a transparent fixed approximate likelihood target with a well-defined gradient. It is not the non-Gaussian density approximation lane; that role belongs to squared tensor-train filtering. The next section therefore pins down the exact state-space problem to which this deterministic Gaussian-surrogate lane is being applied, so that the later sparse-grid, gradient, and comparison arguments refer to one common filtering target.

6 State-Space Model And Exact Filtering Recursion

Use the additive-noise state-space model

$$X_t = f_\theta(X_{t-1}) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_\theta), \quad (6.1)$$

$$Y_t = h_\theta(X_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_\theta), \quad (6.2)$$

with $X_t \in \mathbb{R}^{n_x}$, $Y_t \in \mathbb{R}^{n_y}$, and parameter $\theta \in \mathbb{R}^p$. The exact Bayesian prediction and update are

$$p_\theta(x_t \mid y_{1:t-1}) = \int p_\theta(x_t \mid x_{t-1}) p_\theta(x_{t-1} \mid y_{1:t-1}) dx_{t-1}, \quad (6.3)$$

$$p_\theta(x_t \mid y_{1:t}) = \frac{g_\theta(y_t \mid x_t) p_\theta(x_t \mid y_{1:t-1})}{\int g_\theta(y_t \mid x_t) p_\theta(x_t \mid y_{1:t-1}) dx_t}. \quad (6.4)$$

These are the conceptual filtering equations used by Gaussian approximation filters in Jia, Xin, and Cheng [1, Section 2]. In nonlinear models, the integrals in (6.3) and (6.4) usually do not close in finite-dimensional form. Fixed SGQF replaces those exact densities by a Gaussian moment projection.

6.1 Report-Wide Object Map And Implementation Conventions

The report uses the following principal objects throughout the value and gradient paths. An implementation worker should treat this table as the core object inventory for the note.

symbol	meaning	shape
$\xi^{(r)}$	standardized sparse-grid node in $\mathbb{R}^{n_x}$	n_x
w_r	accumulated signed weight after duplicate merge	scalar
m_t, P_t	carried Gaussian mean and covariance after update	$n_x, n_x \times n_x$
m_t^-, P_t^-	predictive Gaussian mean and covariance before update	$n_x, n_x \times n_x$
C_t, C_t^-	covariance factors with $P_t = C_t C_t^\top$, $P_t^- = C_t^- (C_t^-)^\top$	$n_x \times n_x$
$\chi_{t-1}^{(r)}$	previous-state physical cloud point	n_x
$a_t^{(r)}$	transition image of $\chi_{t-1}^{(r)}$	n_x
$\chi_t^{(r)}$	predictive physical cloud point	n_x
$z_t^{(r)}$	observation image of $\chi_t^{(r)}$	n_y
$\bar{z}_t$	predicted observation mean	n_y
S_t	innovation covariance	$n_y \times n_y$
$C_{xz,t}$	state-observation cross-covariance	$n_x \times n_y$
v_t	innovation residual $y_t - \bar{z}_t$	n_y
K_t	Gaussian projection gain	$n_x \times n_y$
$\hat{\ell}_t^{\text{FSGQ}}$	one-step deterministic approximate log-likelihood increment	scalar

Three implementation conventions are fixed report-wide.

1. **Factor convention.** Every covariance factor is the lower triangular Cholesky factor with positive diagonal, written $C(P) = \text{chol}(P)$, whenever the declared branch is valid.
2. **Solve convention.** Formulas may use S_t^{-1} for compact mathematics, but the implementation contract is to compute all such actions by linear solves against the declared Cholesky factor of S_t , not by forming an explicit inverse.

3. **Noise convention.** Additive process noise enters analytically through the covariance term Q_θ in predictive moments; additive observation noise enters analytically through R_θ in the innovation covariance. This report does not use state augmentation for these additive noises.

When floating-point accumulation produces a covariance that is not exactly symmetric, the report assumes explicit symmetrization by $(P + P^\top)/2$ before branch checking. If the resulting matrix is not positive definite on the declared Cholesky branch, the branch is vetoed rather than silently repaired. Any jitter, clipping, or eigenvalue flooring policy would define a different scalar than the one studied here.

7 Gaussian Projection From Moments

Assume the predictive state is represented by

$$X_t^- \sim \mathcal{N}(m_t^-, P_t^-). \quad (7.1)$$

Let

$$Z_t = h_\theta(X_t^-) \quad (7.2)$$

be the noiseless predicted observation. The Gaussian projection needs

$$\bar{z}_t = \mathbb{E}[Z_t], \quad (7.3)$$

$$S_t = R_\theta + \mathbb{E}[(Z_t - \bar{z}_t)(Z_t - \bar{z}_t)^\top], \quad (7.4)$$

$$C_{xz,t} = \mathbb{E}[(X_t^- - m_t^-)(Z_t - \bar{z}_t)^\top]. \quad (7.5)$$

Given these moments, define

$$v_t = y_t - \bar{z}_t, \quad K_t = C_{xz,t} S_t^{-1}, \quad (7.6)$$

$$m_t = m_t^- + K_t v_t, \quad P_t = P_t^- - K_t S_t K_t^\top. \quad (7.7)$$

Proposition 1 (Affine projection). *Among estimators of X_t^- of the form $a + BY_t$, the minimum mean-square estimator under the moments in (7.3)–(7.5) uses $B = C_{xz,t} S_t^{-1}$. Storing the resulting mean and covariance as a Gaussian gives (7.7).*

Proof. Center $\tilde{X} = X_t^- - m_t^-$ and $\tilde{Y} = Y_t - \bar{z}_t$. The squared-error objective for $m_t^- + B\tilde{Y}$ is

$$J(B) = \mathbb{E}\|\tilde{X} - B\tilde{Y}\|^2 = \text{tr}(P_t^-) - 2\text{tr}(BC_{xz,t}^\top) + \text{tr}(BS_t B^\top). \quad (7.8)$$

Differentiating gives $\nabla_B J = -2C_{xz,t} + 2BS_t$. If S_t is nonsingular, the stationary point is $B = C_{xz,t} S_t^{-1}$. Substitution gives the residual covariance $P_t^- - C_{xz,t} S_t^{-1} C_{xz,t}^\top$, which equals (7.7). $\square$

The approximation has two layers. First, the moments (7.3)–(7.5) may be approximated by quadrature. Second, the resulting posterior information is compressed to one Gaussian.

7.1 How This Matches The Jia–Xin–Cheng Gaussian Approximation Block

Jia, Xin, and Cheng first write the Gaussian approximation filter before they introduce sparse grids [1, Section 2.2]. In their additive-noise setting, the prediction moment equations have the same structure as

$$m_t^- = \mathbb{E}[f_\theta(X_{t-1})], \quad (7.9)$$

$$P_t^- = Q_\theta + \mathbb{E}[(f_\theta(X_{t-1}) - m_t^-)(f_\theta(X_{t-1}) - m_t^-)^\top], \quad (7.10)$$

where $X_{t-1} \sim \mathcal{N}(m_{t-1}, P_{t-1})$ under the carried Gaussian approximation. The update moment equations have the structure

$$\bar{z}_t = \mathbb{E}[h_\theta(X_t^-)], \quad (7.11)$$

$$S_t = R_\theta + \mathbb{E}[(h_\theta(X_t^-) - \bar{z}_t)(h_\theta(X_t^-) - \bar{z}_t)^\top], \quad (7.12)$$

$$C_{xz,t} = \mathbb{E}[(X_t^- - m_t^-)(h_\theta(X_t^-) - \bar{z}_t)^\top]. \quad (7.13)$$

Then the Gaussian projection update is

$$K_t = C_{xz,t} S_t^{-1}, \quad (7.14)$$

$$m_t = m_t^- + K_t(y_t - \bar{z}_t), \quad (7.15)$$

$$P_t = P_t^- - K_t S_t K_t^\top. \quad (7.16)$$

The sparse-grid part of the paper answers only how to approximate the expectations in (7.9)–(7.13). It does not change the fact that the filter stores (m_t, P_t) . In BayesFilter notation, the replacement is

$$\mathbb{E}[F(\xi)] \rightsquigarrow A_{b,L}[F] \rightsquigarrow \sum_{r=1}^{M_{b,L}} w_r F(\xi^{(r)}). \quad (7.17)$$

For prediction,

$$F_{\text{pred}}(\xi) = f_\theta(m_{t-1} + C_{t-1}\xi),$$

and for observation,

$$F_{\text{obs}}(\xi) = h_\theta(m_t^- + C_t^- \xi).$$

This is the exact point where the source SGQF construction enters the filter. Everything before (7.17) is Gaussian projection; everything after it is fixed sparse-grid implementation of the needed Gaussian expectations. Before formalizing the quadrature machinery, it helps to see two small motivating cases that make the computational problem concrete.

8 Why Quadrature Appears: Two Motivating Toy Cases

The next two sections become easier to read if the reader first sees the two questions they are trying to answer:

1. what integral must be approximated once a nonlinear Gaussian moment is needed, and
2. why naive tensor-product point placement becomes expensive as dimension grows.

8.1 Toy case A: one nonlinear Gaussian moment

Start with a scalar Gaussian input and a simple nonlinear observation map. Let

$$X \sim \mathcal{N}(m, P), \quad h(X) = X^2. \quad (8.1)$$

Suppose we want the Gaussian-projection observation mean

$$\bar{z} = \mathbb{E}[h(X)] = \mathbb{E}[X^2]. \quad (8.2)$$

Write $X = m + C\xi$ with $\xi \sim \mathcal{N}(0, 1)$ and $C^2 = P$. Then

$$\bar{z} = \int_{\mathbb{R}} (m + C\xi)^2 \phi(\xi) d\xi. \quad (8.3)$$

This is the first key point: once the state has been standardized, the moment problem becomes a standard-normal weighted integral. The quadrature rule is therefore not an extra modeling choice layered on top of nowhere. It is a numerical approximation to the Gaussian expectation in (8.3).

In this scalar example, a three-point symmetric rule already reproduces the first few Gaussian moments exactly, which is why rules such as $\{-\sqrt{3}, 0, \sqrt{3}\}$ appear naturally later. The purpose of the next section is to formalize this one-dimensional Gaussian rule.

8.2 Toy case B: a two-dimensional Gaussian placement picture

Now move from one coordinate to two. Use a simple two-dimensional Gaussian state,

$$X \sim \mathcal{N}(m, P), \quad X \in \mathbb{R}^2, \quad (8.4)$$

and imagine either a two-dimensional linear-Gaussian state-space step or a slightly nonlinear observation moment built from that same Gaussian input. The point of this example is geometric rather than inferential: it shows how the same standardized cloud is placed and why tensor products grow quickly.

Let $X = m + C\xi$ with $\xi \sim \mathcal{N}(0, I_2)$. If each standardized coordinate uses the three-point rule $\{-\sqrt{3}, 0, \sqrt{3}\}$, then the direct tensor product uses all pairs

$$(\xi_1, \xi_2) \in \{-\sqrt{3}, 0, \sqrt{3}\} \times \{-\sqrt{3}, 0, \sqrt{3}\}, \quad (8.5)$$

so already in two dimensions the rule has $3^2 = 9$ points. In three dimensions the same construction has $3^3 = 27$ points; in ten dimensions it has 3^{10} points. This is the second key point: the tensor-product rule is simple and transparent, but its point count grows too quickly in higher dimension.

Sparse-grid construction tries to keep the same standardized-coordinate logic while avoiding the full tensor explosion. It does that by combining lower-level tensor rules with signed coefficients, then merging duplicate standardized nodes. The next two sections formalize that construction.

9 One-Dimensional Gaussian Quadrature And Tensor Products

The scalar toy case above shows why a Gaussian expectation becomes a standard-normal weighted integral after standardization. This section now makes that one-dimensional rule explicit, and then shows how the same rule is lifted to multiple coordinates by tensor products.

How to read this section with the toy cases in mind. Toy case A should be kept in mind whenever a one-dimensional rule $I_\ell[\psi]$ appears: there the integrand is the scalar nonlinear moment in (8.3). Toy case B should be kept in mind whenever a tensor product appears: there the integrand lives on two standardized coordinates and the point set is the direct Cartesian product in (8.5). The formal notation below is therefore not new mathematics disconnected from the examples; it is the abstract version of those two simple cases.

Let $\xi \sim \mathcal{N}(0, 1)$. A one-dimensional standard-normal quadrature rule at level ℓ is a set of nodes and weights

$$I_\ell[\psi] = \sum_{a=1}^{s_\ell} \omega_{\ell,a} \psi(\zeta_{\ell,a}) \approx \int_{\mathbb{R}} \psi(\xi) \phi(\xi) d\xi. \quad (9.1)$$

In Toy case A, the role of ψ is played by the scalar nonlinear integrand

$$\psi(\xi) = (m + C\xi)^2, \quad (9.2)$$

so the one-dimensional rule simply says: replace the standard-normal integral in (8.3) by a weighted sum of a few standardized abscissae. This is the simplest possible picture of Gaussian quadrature in the present note.

For a multi-index $\mathbf{i} = (i_1, \dots, i_b)$, the corresponding tensor rule is

$$I_{\mathbf{i}}[F] = (I_{i_1} \otimes \dots \otimes I_{i_b})[F]. \quad (9.3)$$

Its nodes and weights are

$$\xi^{\mathbf{i},\alpha} = (\zeta_{i_1,\alpha_1}, \dots, \zeta_{i_b,\alpha_b}), \quad (9.4)$$

$$w^{\mathbf{i},\alpha} = \prod_{j=1}^b \omega_{i_j,\alpha_j}, \quad \alpha_j \in \{1, \dots, s_{i_j}\}. \quad (9.5)$$

In Toy case B with two coordinates and the same three-point rule in each coordinate, this abstract tensor product is exactly the nine-point Cartesian product already listed in (8.5). The notation $\xi^{\mathbf{i},\alpha}$ is just the formal way to say “take one selected standardized point from coordinate 1 and one selected standardized point from coordinate 2, then pair them.”

It helps to see the 2D case written out explicitly. If both coordinates use the three-point rule $\{-\sqrt{3}, 0, \sqrt{3}\}$ with weights $\{1/6, 2/3, 1/6\}$, then the tensor-product rule places the nine points

$$(-\sqrt{3}, -\sqrt{3}), (-\sqrt{3}, 0), (-\sqrt{3}, \sqrt{3}), (0, -\sqrt{3}), (0, 0), (0, \sqrt{3}), (\sqrt{3}, -\sqrt{3}), (\sqrt{3}, 0), (\sqrt{3}, \sqrt{3}), \quad (9.6)$$

with weights given by products of the one-dimensional weights. For example, $(0, 0)$ has weight $(2/3)(2/3) = 4/9$, while $(\sqrt{3}, -\sqrt{3})$ has weight $(1/6)(1/6) = 1/36$. Nothing mysterious has happened yet: tensor products just place all combinations of one-dimensional points.

A full tensor-product rule is simple but expensive: if every coordinate uses s nodes, then $M = s^b$. The two-dimensional toy case above already shows this growth geometrically. With three points per coordinate, two dimensions use nine points, three dimensions use twenty-seven points, and ten dimensions use 3^{10} points. Jia, Xin, and Cheng introduce the sparse-grid combination to reduce this point count for fixed accuracy level [1, Section 3].

10 Sparse-Grid Rule Reconstructed In Source Order

The sparse-grid construction answers a specific question raised by the toy cases. The scalar example shows why nonlinear Gaussian moments become standard-normal weighted integrals after standardization. The two-dimensional example shows why naive tensor products grow quickly even when the placement logic is simple. The purpose of sparse-grid combination is to keep the standardized-coordinate moment logic while reducing the point count relative to the full tensor product.

How to read this section with the toy cases in mind. Keep returning mentally to Toy case B. There, the direct tensor-product rule uses the full nine-point grid in (9.6). Sparse-grid combination should be read as a structured way to approximate the same Gaussian moment target without paying for every point of the full tensor product at every level. The formal index sets,

coefficients, and duplicate-node merges below are not separate ideas; they are the bookkeeping machinery that turns that intuition into a reproducible fixed cloud.

Suppose the filter needs a Gaussian expectation

$$\mathcal{I}_b[F] = \int_{\mathbb{R}^b} F(\xi) \phi_b(\xi) d\xi, \quad \phi_b(\xi) = (2\pi)^{-b/2} \exp(-\|\xi\|^2/2). \quad (10.1)$$

In Toy case A, this is just the scalar standard-normal integral in (8.3). In Toy case B with two coordinates, it is the two-dimensional Gaussian-weighted integral approximated by the nine-point tensor grid in (9.6). The higher-dimensional notation $\mathcal{I}_b[F]$ is therefore the common umbrella covering those simple examples and the later filtering moments.

A full tensor rule approximates $\mathcal{I}_b[F]$ by evaluating F on every Cartesian product of one-dimensional nodes. If each coordinate uses s points, the point count is s^b . Jia, Xin, and Cheng replace that full tensor rule by a signed linear combination of lower-level tensor rules [1, Section 3.1]. The signs are not an implementation accident. They are inclusion-exclusion coefficients: low-level tensor rules overlap, and overlapping contributions must be subtracted before higher-level contributions are added.

10.1 The Jia–Xin–Cheng Index Sets

Jia, Xin, and Cheng group tensor-product levels by an auxiliary integer q [1, Eq. 27]. In BayesFilter notation, for dimension b ,

$$N_q^b = \left\{ \mathbf{i} = (i_1, \dots, i_b) \in \mathbb{N}^b : \sum_{j=1}^b i_j = b + q \right\}, \quad q \geq 0, \quad (10.2)$$

and $N_q^b = \emptyset$ for $q < 0$. A member $\mathbf{i} \in N_q^b$ says: use the one-dimensional rule I_{i_j} in coordinate j , then take their tensor product. The larger q is, the more coordinates can have levels above one.

For a declared sparse-grid level L , the source sums over

$$q = L - b, L - b + 1, \dots, L - 1.$$

Since N_q^b is empty when $q < 0$, this formula also covers the case $L < b$. If $\mathbf{i} \in N_q^b$, then

$$|\mathbf{i}| = \sum_{j=1}^b i_j = b + q. \quad (10.3)$$

Thus the same index set can be written without q as

$$\mathcal{A}_{b,L} = \{\mathbf{i} \in \mathbb{N}^b : L \leq |\mathbf{i}| \leq L + b - 1\}, \quad |\mathbf{i}| = \sum_{j=1}^b i_j. \quad (10.4)$$

This is exactly the band used by the implementable fixed-cloud construction in this note.

To keep the 2D toy case in view, imagine $b = 2$. Then the active band says: not every tensor-product level is kept. Instead, only a narrow range of total multi-index levels is retained. This is the first place where sparse-grid construction departs from the full nine-point tensor-product picture. The rule is no longer “use every product level equally”; it is “keep a structured band of tensor levels whose total index lies in a narrow admissible range.”

10.2 The Smolyak Coefficients

In the source formula, the coefficient attached to the q -slice is

$$(-1)^{L-1-q} \binom{b-1}{L-1-q}.$$

Using $q = |\mathbf{i}| - b$ from (10.3), this becomes

$$c_{\mathbf{i}} = (-1)^{L+b-1-|\mathbf{i}|} \binom{b-1}{L+b-1-|\mathbf{i}|}. \quad (10.5)$$

Therefore the sparse-grid quadrature approximation is

$$A_{b,L}[F] = \sum_{\mathbf{i} \in \mathcal{A}_{b,L}} c_{\mathbf{i}} I_{\mathbf{i}}[F]. \quad (10.6)$$

Equation (10.6) is the BayesFilter form of Jia et al. [1, Eq. 26–29]. The coefficient (10.5) can be negative. That fact later explains why the filter must check covariance matrices instead of assuming every weighted second-moment sum is automatically positive semidefinite.

In the 2D toy case, these signed coefficients are the formal reason the sparse rule is not just a smaller positive-weight subset of the nine-point tensor grid. It is a signed combination of several lower-level tensor rules. Some points are added more than once, some overlap, and some overlapping contributions must be subtracted. This is exactly why duplicate-node merging matters later: after the signed combination is expanded into explicit tensor points, the same standardized location can arise from more than one tensor component.

10.3 Univariate Moment Matching

The one-dimensional rules I_{ℓ} must be chosen before (10.6) becomes a concrete point set. Jia, Xin, and Cheng construct univariate rules by matching moments of a standard normal [1, Section 3.2]. If p_a and ω_a are the one-dimensional nodes and weights, the matching equations are

$$M_j = \int_{-\infty}^{\infty} x^j \phi(x) dx = \sum_{a=1}^{N_u} \omega_a p_a^j, \quad j = 0, 1, \dots, J. \quad (10.7)$$

The moments on the left are explicit:

$$M_j = \begin{cases} 0, & j \text{ odd}, \\ (j-1)(j-3) \cdots 3 \cdot 1, & j \text{ even}. \end{cases} \quad (10.8)$$

For a symmetric three-point rule

$$\{(-a, \omega), (0, \omega_0), (a, \omega)\},$$

the first three nontrivial equations are

$$\omega_0 + 2\omega = 1, \quad (10.9)$$

$$2\omega a^2 = 1, \quad (10.10)$$

$$2\omega a^4 = 3. \quad (10.11)$$

Dividing (10.11) by (10.10) gives $a^2 = 3$. Then $\omega = 1/6$ and $\omega_0 = 2/3$. This derivation is why the toy cloud below uses nodes $\{-\sqrt{3}, 0, \sqrt{3}\}$ with weights $\{1/6, 2/3, 1/6\}$. The same moment-matching logic extends to the higher level symmetric rules discussed by Jia et al. [1, Section 3.2].

This is also the exact place where Toy case A connects back to the formal rule. The scalar nonlinear moment in (8.3) needs a one-dimensional Gaussian rule. The three-point rule just derived is the first concrete example of such a rule. Toy case B then reuses the same one-dimensional rule coordinatewise before tensor products or sparse-grid combinations are even considered.

10.4 From Formula To Point Cloud

Equation (10.6) is still not an implementation object. Each tensor rule $I_{\mathbf{i}}$ has points and weights

$$\xi^{\mathbf{i},\alpha} = (\zeta_{i_1,\alpha_1}, \dots, \zeta_{i_b,\alpha_b}), \quad (10.12)$$

$$w^{\mathbf{i},\alpha} = \prod_{j=1}^b \omega_{i_j,\alpha_j}, \quad \alpha_j \in \{1, \dots, s_{i_j}\}. \quad (10.13)$$

The sparse-grid contribution of this tensor point is

$$c_{\mathbf{i}} w^{\mathbf{i},\alpha}. \quad (10.14)$$

The same standardized point can appear in several tensor components. Jia, Xin, and Cheng's Algorithm 1 says to add a new point if it has not appeared before, and otherwise update the old weight [1, Algorithm 1]. The mathematical object is therefore the accumulated dictionary

$$D[\xi] = \sum_{\substack{\mathbf{i} \in \mathcal{A}_{b,L} \\ \alpha: \xi^{\mathbf{i},\alpha} = \xi}} c_{\mathbf{i}} w^{\mathbf{i},\alpha}. \quad (10.15)$$

After merging duplicate standardized nodes, enumerate the nonzero entries as

$$\mathcal{C}_{b,L} = \{(\xi^{(r)}, w_r)\}_{r=1}^{M_{b,L}}. \quad (10.16)$$

This report adopts the following deterministic cloud-construction convention.

1. Enumerate multi-indices $\mathbf{i} \in \mathcal{A}_{b,L}$ in lexicographic order.
2. Within each tensor rule $I_{\mathbf{i}}$, enumerate the local index tuples α lexicographically.
3. For each candidate standardized node $\xi^{\mathbf{i},\alpha}$, search the current dictionary for an existing node ξ satisfying

$$\|\xi^{\mathbf{i},\alpha} - \xi\|_{\infty} \leq \tau_{\text{merge}} := 10^{-12} \max\{1, \|\xi\|_{\infty}\}. \quad (10.17)$$

If such a node exists, accumulate the signed weight $c_{\mathbf{i}} w^{\mathbf{i},\alpha}$ into its stored weight; otherwise add a new dictionary entry.

4. Remove entries whose accumulated absolute weight is below the declared numerical-zero threshold.
5. Sort the surviving merged nodes lexicographically and store the final cloud in that order.

For the fixed BayesFilter scalar, $\mathcal{C}_{b,L}$, the univariate rules, the merge tolerance, the zero-weight rule, the sorting convention, and the accumulated signed weights are part of the target. If two implementations use the same level L but merge nearly-equal floating-point nodes differently, they have not necessarily evaluated the same scalar.

Seen through Toy case B, this is the final conceptual step. The tensor-product picture starts with explicit point sets such as the nine-point grid in (9.6). Sparse-grid construction then says: combine selected tensor rules with signed coefficients, collect all resulting point contributions in one dictionary, merge repeated standardized nodes, and only then store the final fixed cloud. The cloud used later by the filter is therefore not a raw tensor component and not a symbolic Smolyak formula. It is the merged, sorted, weighted list $\mathcal{C}_{b,L}$ produced after all of that bookkeeping has been completed.

Implementation-facing pseudocode summary. The runtime object produced by this section is a stored cloud builder routine of the form `build_fixed_sparse_grid_cloud((b, L, {Iℓ}, τmerge, τzero))` that returns the sorted merged cloud $\mathcal{C}_{b,L}$, its weight-total diagnostic, and any signed-weight diagnostics used later in validation. The mathematics above is the authoritative definition of that routine.

10.5 Exactness, Point Count, And The UKF Relation

Theorem 3.1 of Jia et al. [1] is a quadrature exactness statement. If each univariate I_ℓ is exact for univariate polynomials through degree $2\ell - 1$, then the sparse-grid rule is exact for multivariate polynomials through the corresponding total degree stated in that theorem. The theorem supports this inference:

$$F \text{ polynomial in the exactness class} \implies A_{b,L}[F] = \mathcal{I}_b[F]. \quad (10.18)$$

It does not support the stronger inference

$$\text{nonlinear posterior is exactly represented by one Gaussian.}$$

FixedSGQF uses (10.18) to justify deterministic moment approximation, not to claim exact Bayesian filtering.

The point-count proposition in Jia et al. [1, Proposition 3.1] also has a precise scope. For fixed level L , the number of sparse-grid points grows polynomially in dimension, with highest order $L - 1$ under the source conditions. The core counting reason is visible from (10.2): in the largest relevant slice, at most $L - 1$ coordinates can have levels above one. The number of ways to choose those coordinates is of order $\binom{b}{L-1}$, which is polynomial in b when L is fixed. This explains the storage advantage over full tensor products. It does not choose the correct level for a hard model.

The nestedness proposition in Jia et al. [1, Proposition 3.2] explains when lower-level point sets are contained in higher-level point sets. For implementation, its practical message is simple: nested univariate rules can reduce the number of distinct points after merging. Non-nested rules can still be used, but the duplicate merge will be weaker and the point count can be larger.

Finally, Jia et al. [1, Theorem 3.2] shows that the unscented transform point set appears as a level-2 sparse-grid member under the source's choice of one-point and three-point univariate rules. Here this theorem is used as a conceptual bridge. It tells a reader that SGQF is not a mysterious new filtering principle; it is a systematic sparse-grid way to build deterministic Gaussian moment points. The selected BayesFilter object remains the fixed cloud (10.16).

10.6 A Deterministic Level Ladder Before Failure

For implementation work, "choose a level" is not enough. Use a declared deterministic ladder:

$$L \in \{2, 3, 4, \dots, L_{\max}\}, \quad (10.19)$$

with a point budget $M_{\max}$, a covariance branch policy, and a validation set of pilot parameter values Θ_{pilot} . For each level, build $\mathcal{C}_{b,L}$, record $M_{b,L}$, and run the construction diagnostics and pilot filtering diagnostics. The first level that satisfies

$$M_{b,L} \leq M_{\max}, \quad (10.20)$$

$$\min_{t, \theta \in \Theta_{\text{pilot}}} \lambda_{\min}(S_t(\theta)) \geq \epsilon_S, \quad (10.21)$$

$$\max_{\theta \in \Theta_{\text{pilot}}} \left| \widehat{\ell}_T^{(L)}(\theta) - \widehat{\ell}_T^{(L-1)}(\theta) \right| \leq \epsilon_\ell \quad \text{for } L > 2 \quad (10.22)$$

is eligible for a fixed-branch likelihood. If no level satisfies the ladder, FixedSGQF returns a design failure rather than silently using a convenient grid. The numerical constants ϵ_S and ϵ_ℓ are diagnostic thresholds, not mathematical proof parameters. A minimal default table is given later in Section 17.3.

11 A Toy Fixed Grid With Duplicate Merging

This section makes the sparse-grid cloud concrete. Take $b = 2$ and $L = 2$. Use

$$I_1 : (0, 1), \quad I_2 : \left\{ (-\sqrt{3}, 1/6), (0, 2/3), (\sqrt{3}, 1/6) \right\}, \quad (11.1)$$

which is the familiar three-point standard-normal rule at level 2. The band $\mathcal{A}_{2,2}$ contains

$$(1, 1), \quad (1, 2), \quad (2, 1), \quad (11.2)$$

with coefficients

$$c_{(1,1)} = -1, \quad c_{(1,2)} = 1, \quad c_{(2,1)} = 1. \quad (11.3)$$

Before merging, the node $(0, 0)$ appears three times. Its accumulated weight is

$$-1 + \frac{2}{3} + \frac{2}{3} = \frac{1}{3}. \quad (11.4)$$

The four axis nodes each receive weight $1/6$. The merged cloud is

node ξ	accumulated weight	source components
$(0, 0)$	$1/3$	$(1, 1), (1, 2), (2, 1)$
$(-\sqrt{3}, 0)$	$1/6$	$(2, 1)$
$(\sqrt{3}, 0)$	$1/6$	$(2, 1)$
$(0, -\sqrt{3})$	$1/6$	$(1, 2)$
$(0, \sqrt{3})$	$1/6$	$(1, 2)$

The weights sum to one:

$$\frac{1}{3} + 4 \cdot \frac{1}{6} = 1. \quad (11.5)$$

This toy cloud illustrates two implementation rules that are easy to get wrong. First, duplicate nodes must be accumulated before the cloud is used. Second, signed sparse-grid coefficients are applied before accumulation; a future level or a different univariate family can produce negative accumulated weights, so covariance matrices require explicit diagnostics.

Algorithmically, this toy example should be read as a dictionary update. Start from an empty map from standardized nodes to accumulated weights. Insert the contribution of each tensor component in (11.2) with its signed coefficient from (11.3). The origin node $(0,0)$ receives three signed contributions and is stored only once with the final accumulated weight from (11.4). The resulting five-node dictionary is exactly the object that is later reused by both the value and gradient paths; no tensor-component provenance is needed after the merge is complete.

12 FixedSGQF Filtering Value Path

Fix a standardized sparse-grid cloud

$$\mathcal{C} = \{(\xi^{(r)}, w_r)\}_{r=1}^M, \quad \xi^{(r)} \in \mathbb{R}^{n_x}, \quad w_r \in \mathbb{R}. \quad (12.1)$$

The cloud is frozen before likelihood evaluation. The value path has two quadrature stages at each time.

What this section gives the implementer. This section is the authoritative one-step value recursion. A value routine constructed from this note should take the current carried Gaussian, current observation, fixed cloud, and declared model objects as input, then return either an accepted one-step increment with updated (m_t, P_t) or a branch failure record.

12.1 Prediction Stage

Let $P_{t-1} = C_{t-1}C_{t-1}^\top$ on the declared square-root branch. Place the previous Gaussian cloud:

$$\chi_{t-1}^{(r)} = m_{t-1} + C_{t-1}\xi^{(r)}. \quad (12.2)$$

Push points through the transition:

$$a_t^{(r)} = f_\theta(\chi_{t-1}^{(r)}). \quad (12.3)$$

The predicted Gaussian is

$$m_t^- = \sum_{r=1}^M w_r a_t^{(r)}, \quad (12.4)$$

$$P_t^- = Q_\theta + \sum_{r=1}^M w_r (a_t^{(r)} - m_t^-)(a_t^{(r)} - m_t^-)^\top. \quad (12.5)$$

After (12.5), the implementation replaces P_t^- by $(P_t^- + P_t^{-\top})/2$ and checks the declared positive-definite branch. For the differentiable lane in this note, there is no adaptive jitter, floor, clipping, pivot, or eigenvalue repair inside the scalar. If the check fails, the value path returns a veto rather than quietly changing the scalar.

12.2 Observation Stage

Factor $P_t^- = C_t^-(C_t^-)^\top$ on the declared branch and place the predictive cloud:

$$\chi_t^{(r)} = m_t^- + C_t^- \xi^{(r)}. \quad (12.6)$$

Evaluate the observation map:

$$z_t^{(r)} = h_\theta(\chi_t^{(r)}). \quad (12.7)$$

Then compute

$$\bar{z}_t = \sum_{r=1}^M w_r z_t^{(r)}, \quad (12.8)$$

$$\Delta_t^{(r)} = z_t^{(r)} - \bar{z}_t, \quad (12.9)$$

$$S_t = R_\theta + \sum_{r=1}^M w_r \Delta_t^{(r)} \Delta_t^{(r)\top}, \quad (12.10)$$

$$C_{xz,t} = \sum_{r=1}^M w_r (\chi_t^{(r)} - m_t^-) \Delta_t^{(r)\top}. \quad (12.11)$$

The innovation and log likelihood are

$$v_t = y_t - \bar{z}_t, \quad (12.12)$$

$$\hat{\ell}_t^{\text{FSGQ}} = -\frac{1}{2} \left\{ \log \det S_t + v_t^\top S_t^{-1} v_t + n_y \log(2\pi) \right\}. \quad (12.13)$$

If S_t is not positive definite on the declared branch, the value path returns a veto. If it passes, the Gaussian projection update is

$$K_t = C_{xz,t} S_t^{-1}, \quad (12.14)$$

$$m_t = m_t^- + K_t v_t, \quad (12.15)$$

$$P_t = P_t^- - K_t S_t K_t^\top. \quad (12.16)$$

Value-path pseudocode summary. The runtime order defined by this section is:

1. symmetrize and factor P_{t-1} ,
2. place previous-state points and propagate through f_θ ,
3. form (m_t^-, P_t^-) and veto if the predictive branch fails,
4. factor P_t^- , place predictive points, and propagate through h_θ ,
5. form $\bar{z}_t$, S_t , and $C_{xz,t}$, and veto if the innovation branch fails,
6. form v_t , $\hat{\ell}_t^{\text{FSGQ}}$, K_t , m_t , and P_t , then veto if the carried covariance branch fails,
7. otherwise return the one-step increment, updated carried Gaussian, and any diagnostics cached for derivative reuse.

This pseudocode summary does not replace the equations above; it states their required execution order.

13 A Worked One-Step Example With A Numeric Oracle

This section gives one compact example that can be read two ways. First, it is small enough for a chair to teach back without reconstructing the notation from scratch. Second, it gives an implementation engineer a concrete one-step oracle against which a first prototype can be checked.

Use the one-dimensional transition-observation model

$$X_0 \sim \mathcal{N}(m_0, P_0), \quad m_0 = 0.5, \quad P_0 = 1, \quad (13.1)$$

$$X_1 = f_\beta(X_0) + \eta_1, \quad f_\beta(x) = \beta x, \quad \eta_1 \sim \mathcal{N}(0, Q), \quad (13.2)$$

$$Y_1 = h_\theta(X_1) + \varepsilon_1, \quad h_\theta(x) = x^2, \quad \varepsilon_1 \sim \mathcal{N}(0, R), \quad (13.3)$$

with

$$Q = 0.25, \quad R = 1, \quad y_1 = 2. \quad (13.4)$$

Differentiate with respect to the single parameter component β . For this chosen component, the transition depends on β but the observation map does not, so

$$\partial_\beta h_\theta(x) = 0. \quad (13.5)$$

Evaluate all quantities at $\beta = 1$, and use the positive scalar Cholesky branch $C(P) = \sqrt{P}$ with no repair. Because the example yields $P_1^- = 5/4 > 0$ and $S_1 = 43/8 > 0$, both the value and derivative lanes stay on that same declared branch. Use the three-point standard-normal rule

$$\mathcal{C}_{1,2} = \left\{ \left(-\sqrt{3}, \frac{1}{6} \right), \left(0, \frac{2}{3} \right), \left(\sqrt{3}, \frac{1}{6} \right) \right\}. \quad (13.6)$$

The same three-node cloud is reused without change in both the value and derivative calculations, so the example can be taught back as one declared branch with one declared cloud across both lanes.

Value path. Since $C_0 = \sqrt{P_0} = 1$, the previous-state placed points are

$$\chi_0^{(r)} = m_0 + C_0 \xi^{(r)} \in \{0.5 - \sqrt{3}, 0.5, 0.5 + \sqrt{3}\}. \quad (13.7)$$

At $\beta = 1$, the transition images satisfy

$$a_1^{(r)} = f_\beta(\chi_0^{(r)}) = \beta \chi_0^{(r)} = \chi_0^{(r)}. \quad (13.8)$$

The predicted Gaussian moments are therefore

$$m_1^- = \sum_r w_r a_1^{(r)} = 0.5, \quad P_1^- = Q + \sum_r w_r (a_1^{(r)} - m_1^-)^2 = 1.25 = \frac{5}{4}. \quad (13.9)$$

The declared Cholesky factor is

$$C_1^- = \text{chol}(P_1^-) = \sqrt{1.25} = \frac{\sqrt{5}}{2}. \quad (13.10)$$

Hence the predictive placed points are

$$\chi_1^{(r)} = m_1^- + C_1^- \xi^{(r)} \in \left\{ 0.5 - \frac{\sqrt{15}}{2}, 0.5, 0.5 + \frac{\sqrt{15}}{2} \right\}. \quad (13.11)$$

Passing those points through $h_\theta(x) = x^2$ gives

$$z_1^{(r)} \in \{2.0635083269, 0.25, 5.9364916731\}. \quad (13.12)$$

The centered predictive and observation values are

$$d_1^{(r)} = a_1^{(r)} - m_1^- \in \{-\sqrt{3}, 0, \sqrt{3}\}, \quad (13.13)$$

and

$$\Delta_1^{(r)} = z_1^{(r)} - \bar{z}_1 \in \{0.5635083269, -1.25, 4.4364916731\}. \quad (13.14)$$

The observation moments and Gaussian update objects are

$$\bar{z}_1 = \sum_r w_r z_1^{(r)} = 1.5 = \frac{3}{2}, \quad (13.15)$$

$$S_1 = R + \sum_r w_r (z_1^{(r)} - \bar{z}_1)^2 = 5.375 = \frac{43}{8}, \quad (13.16)$$

$$C_{xz,1} = \sum_r w_r (\chi_1^{(r)} - m_1^-)(z_1^{(r)} - \bar{z}_1) = 1.25 = \frac{5}{4}, \quad (13.17)$$

$$v_1 = y_1 - \bar{z}_1 = 0.5 = \frac{1}{2}, \quad (13.18)$$

$$K_1 = C_{xz,1} S_1^{-1} = \frac{10}{43} \approx 0.2325581395, \quad (13.19)$$

$$m_1 = m_1^- + K_1 v_1 = \frac{53}{86} \approx 0.6162790698, \quad (13.20)$$

$$P_1 = P_1^- - K_1 S_1 K_1^\top = \frac{165}{172} \approx 0.9593023256. \quad (13.21)$$

The one-step deterministic approximate log likelihood is

$$\hat{\ell}_1^{\text{FSGQ}} = -\frac{1}{2} \{ \log S_1 + v_1^2 S_1^{-1} + \log(2\pi) \} \approx -1.7830736342. \quad (13.22)$$

Same-branch derivative with respect to β . The same cloud and the same positive scalar Cholesky branch are reused in the derivative lane. This example is chosen so that the parameter enters only through the transition, not through the observation map, which is why $\partial_\beta h_\theta(x) = 0$. Because m_0 and P_0 are fixed in β , the previous-state placed points satisfy

$$\dot{\chi}_0^{(r)} = 0. \quad (13.23)$$

From (15.11),

$$\dot{a}_1^{(r)} = D_x f_\beta(\chi_0^{(r)}) \dot{\chi}_0^{(r)} + \partial_\beta f_\beta(\chi_0^{(r)}) = \chi_0^{(r)} \in \{0.5 - \sqrt{3}, 0.5, 0.5 + \sqrt{3}\}. \quad (13.24)$$

Hence

$$\dot{m}_1^- = \sum_r w_r \dot{a}_1^{(r)} = m_0 = 0.5, \quad \dot{d}_1^{(r)} = \dot{a}_1^{(r)} - \dot{m}_1^- \in \{-\sqrt{3}, 0, \sqrt{3}\}, \quad (13.25)$$

and

$$\dot{P}_1^- = \partial_\beta(\beta^2 P_0 + Q) = 2. \quad (13.26)$$

Because $C_1^- = \sqrt{P_1^-}$, the declared Cholesky derivative is

$$\dot{C}_1^- = \frac{\dot{P}_1^-}{2C_1^-} = \frac{2}{\sqrt{5}} \approx 0.8944271910. \quad (13.27)$$

Therefore the predictive point sensitivities are

$$\dot{\chi}_1^{(r)} = \dot{m}_1^- + \dot{C}_1^- \xi^{(r)} \in \{-1.0491933385, 0.5, 2.0491933385\}. \quad (13.28)$$

Since $h_\theta(x) = x^2$ and $\partial_\beta h_\theta(x) = 0$,

$$\dot{z}_1^{(r)} = 2\chi_1^{(r)} \dot{\chi}_1^{(r)} \in \{3.0143149884, 0.5, 9.9856850116\}. \quad (13.29)$$

The centered observation sensitivities are

$$\dot{\Delta}_1^{(r)} = \dot{z}_1^{(r)} - \dot{z}_1 \in \{0.5143149884, -2, 7.4856850116\}. \quad (13.30)$$

The derived moment sensitivities are

$$\dot{z}_1 = 2.5 = \frac{5}{2}, \quad \dot{v}_1 = -2.5 = -\frac{5}{2}, \quad (13.31)$$

$$\dot{S}_1 = 14.5 = \frac{29}{2}, \quad \dot{C}_{xz,1} = 3.25 = \frac{13}{4}. \quad (13.32)$$

Let

$$u_1 = S_1^{-1} v_1 = \frac{4}{43} \approx 0.0930232558. \quad (13.33)$$

The three scalar ingredients in the score formula are

$$\text{tr}(S_1^{-1} \dot{S}_1) = \frac{\dot{S}_1}{S_1} = \frac{116}{43} \approx 2.6976744186, \quad (13.34)$$

$$2\dot{v}_1 u_1 = -\frac{20}{43} \approx -0.4651162791, \quad (13.35)$$

$$-u_1^2 \dot{S}_1 = -\frac{232}{1849} \approx -0.1254732288. \quad (13.36)$$

Then the same-branch one-step score is

$$\dot{\ell}_1^{\text{FSGQ}} = -\frac{1}{2} \left[\frac{\dot{S}_1}{S_1} + 2\dot{v}_1 u_1 - u_1^2 \dot{S}_1 \right] = -\frac{1948}{1849} \approx -1.0535424554. \quad (13.37)$$

For later propagation,

$$\dot{K}_1 = \dot{C}_{xz,1} S_1^{-1} - K_1 \dot{S}_1 S_1^{-1} = -\frac{42}{1849} \approx -0.0227149811, \quad (13.38)$$

so

$$\dot{m}_1 = \dot{m}_1^- + \dot{K}_1 v_1 + K_1 \dot{v}_1 = -\frac{343}{3698} \approx -0.0927528394, \quad (13.39)$$

$$\dot{P}_1 = \dot{P}_1^- - \dot{K}_1 S_1 K_1 - K_1 \dot{S}_1 K_1 - K_1 S_1 \dot{K}_1 = \frac{2353}{1849} \approx 1.2725797729. \quad (13.40)$$

The example makes the report's main limitation concrete. Every quantity above is deterministic and internally checkable on the declared branch, yet the observation law $Y_1 = X_1^2 + \varepsilon_1$ still has the same qualitative ability to generate non-Gaussian posterior structure emphasized earlier in the scalar-cell warning. FixedSGQF therefore gives a transparent Gaussian-surrogate value-and-gradient oracle, not a full non-Gaussian posterior representation.

Implementation checklist for this oracle. A first implementation should be able to reproduce, up to declared floating-point tolerance, at least the following quantities from this section: $m_1^-, P_1^-, \bar{z}_1, S_1, C_{xz,1}, K_1, m_1, P_1, \hat{\ell}_1^{\text{FSGQ}}$, and $\hat{\ell}_1^{\text{FSGQ}}$. This is the compact end-to-end numerical oracle for both the value path and the derivative path.

The next section now states the exact saved-scalar contract that the worked example has instantiated numerically: which structural choices are frozen, which numerical quantities are recomputed, and what it means for value, gradient, and finite differences to refer to the same declared target.

14 The Saved Scalar And Same-Scalar Contract

The derivative in this note is meaningful only for the scalar actually evaluated. This section therefore does two things at once: it freezes the exact structural choices that define the target, and it prepares the reader for why the later derivative chapter must be read as the derivative of that declared target rather than of a changing algorithm. A fixed-SGQF branch is the tuple

$$\begin{aligned} \mathcal{B} = & (y_{1:T}, \text{ observation preprocessing, } m_0(\theta), P_0(\theta), \dot{m}_0, \dot{P}_0, \\ & \mathcal{C}, \text{ one-dimensional rules, duplicate-merge tolerance, zero-weight rule, node ordering,} \\ & C(P) = \text{chol}(P), \text{ symmetrize-then-veto rule, } \epsilon_S, \epsilon_P, \\ & f_\theta, h_\theta, Q_\theta, R_\theta, D_x f_\theta, \partial_i f_\theta, D_x h_\theta, \partial_i h_\theta, \dot{Q}_\theta, \dot{R}_\theta). \end{aligned} \quad (14.1)$$

The fixed scalar is

$$\hat{\ell}_T^{\text{FSGQ}}(\theta; \mathcal{B}) = \sum_{t=1}^T \hat{\ell}_t^{\text{FSGQ}}(\theta; \mathcal{B}), \quad (14.2)$$

where every term is computed by (12.2)–(12.16). The same $\mathcal{B}$ must be used for values, gradients, and finite differences. Changing the grid level, active index set, duplicate-merge tolerance, zero-weight pruning rule, node ordering, Cholesky pivot, observation preprocessing, initial-condition policy, eigenvalue floor, covariance clipping, or point pruning changes the scalar. The derivative formulas below cover the open branch where the symmetrized covariances pass the Cholesky and positive-definite checks without repair. If a jitter, floor, clipping, or pivoting map is desired, it must be declared as a different scalar with its own derivative or stop rule.

Plain-language branch rule. The branch identity freezes structure, not numerical values. When θ changes inside a same-scalar finite-difference check, the moments, gains, and likelihood contributions are recomputed, but they must be recomputed by the same cloud, the same merge/order policy, the same factor family, and the same realized acceptance path through the declared branch logic. This is the operational meaning of “same scalar” for implementation review.

Frozen versus variable objects on a saved branch. The branch record $\mathcal{B}$ freezes structural choices: the cloud, merge/prune/order policy, factor family, veto thresholds, preprocessing policy, and initial-condition policy. It does *not* freeze the numerical values of the moments or filters. Within a fixed branch,

$$\begin{aligned} \text{frozen structural objects: } & \mathcal{C}, \tau_{\text{merge}}, \tau_{\text{zero}}, \text{ ordering, } C(\cdot), \epsilon_S, \epsilon_P, \\ \text{parameter-dependent numerical objects: } & m_t, P_t, C_t, m_t^-, P_t^-, C_t^-, \bar{z}_t, S_t, C_{xz,t}, K_t, \hat{\ell}_t^{\text{FSGQ}}. \end{aligned} \quad (14.3)$$

Thus same-scalar finite differences recompute the parameter-dependent numerical objects by the same declared branch rule; they do not reuse old numerical values. In particular, $m_0(\theta \pm he_i)$ and $P_0(\theta \pm he_i)$ are recomputed by the same initial-condition policy; only the policy is frozen, not the numerical initial values. What must remain unchanged is the structural route by which those values are produced.

Operational branch identity for finite differences. For implementation review, define the branch-identity record

$$\mathfrak{B} = (\mathcal{C}, \tau_{\text{merge}}, \tau_{\text{zero}}, \text{ordering}, \text{preprocessing policy}, \text{initial-condition policy}, \text{symmetrize-then-veto rule}, C(P) = \quad (14.4)$$

The accepted/failure stage-time pattern means the realized branch path through each likelihood contribution: at every time step it records whether the predictive covariance check, innovation covariance check, and carried-covariance check were accepted, and if not, which stage failed first. A central, forward, or backward finite-difference row is branch-valid only if both perturbed evaluations share the same saved structural metadata and the same realized acceptance path as the base evaluation through every likelihood contribution included in the comparison. Equal factor family but different first failure location counts as branch mismatch. Equal first failure location but different accepted-prefix pattern also counts as branch mismatch. If either side changes the accepted stage-time pattern, triggers a different veto location, or uses a different cloud/merge/order/preprocessing/initial-condition specification, the row is branch-invalid and is not interpreted as evidence about the derivative.

15 Analytical Gradient Of The Fixed Scalar

The value path has now fixed the declared scalar and the branch on which that scalar is evaluated. The next task is not to introduce a new target, but to differentiate that same declared target in a way that can still be checked by finite differences. Differentiate with respect to parameter component θ_i . A dot denotes ∂_i , for example $\dot{m}_t = \partial_i m_t$. The data y_t are recorded and do not depend on θ .

Why this derivative is operationally delicate. The one-step scalar at time t depends on objects constructed earlier in the filter: the previous carried Gaussian, the covariance factor used to place the cloud, the nonlinear transition and observation images, the innovation covariance, and the posterior update. The derivative must therefore propagate through the entire recursion, and it is meaningful only when the value and derivative paths stay on the same declared branch.

Current-score objects versus propagation objects. For the current likelihood contribution, the derivative closes once $\dot{v}_t$ and $\dot{S}_t$ are known. The derivative $\dot{C}_{xz,t}$ is not needed to close the current score, but it is needed to form $\dot{K}_t$ and hence to propagate $\dot{m}_t$ and $\dot{P}_t$ to the next time step. A correct implementation must keep these two roles separate.

15.1 The Story Of The Derivative

The derivative is long because the scalar at time t depends on objects created earlier in the filter. This section therefore proceeds in six explicit stages: factor derivative, point sensitivities, predictive moment sensitivities, observation moment sensitivities, innovation score, and posterior sensitivity

propagation. The chain is

$$\begin{aligned} \theta \mapsto (m_{t-1}, P_{t-1}) \mapsto C_{t-1} \mapsto \chi_{t-1}^{(r)} \mapsto a_t^{(r)} \mapsto (m_t^-, P_t^-) \mapsto C_t^- \\ \mapsto \chi_t^{(r)} \mapsto z_t^{(r)} \mapsto (\bar{z}_t, S_t, C_{xz,t}) \mapsto (v_t, K_t, \hat{\ell}_t, m_t, P_t). \end{aligned} \quad (15.1)$$

The sparse-grid nodes and weights do not appear with dots in this chain. They are saved branch objects. Their job is to say where the deterministic moment sum is evaluated, not to move when θ moves. The Gaussian factors C_{t-1} and C_t^- , however, do move with θ , because they are functions of P_{t-1} and P_t^- . That is why the factor derivative is part of the scalar contract.

For readers tracking the derivation operationally, the six stages below can be read as a ledger of which object is introduced, what it depends on, and whether it closes the present score or only prepares the next step.

$$\text{extfactororderivative} \Rightarrow \dot{C}_{t-1}, \dot{C}_t^-, \text{extpredictionstage} \Rightarrow \dot{\chi}_{t-1}^{(r)}, \dot{a}_t^{(r)}, \dot{m}_t^-, \dot{P}_t^-, \text{extobservationstage} \Rightarrow \dot{\chi}_t^{(r)}, \dot{z}_t^{(r)} \quad (15.2)$$

The value path computes the scalar

$$\hat{\ell}_t = -\frac{1}{2} \{ \log \det S_t + v_t^\top S_t^{-1} v_t + n_y \log(2\pi) \}.$$

The gradient path differentiates the same line. It is helpful to separate the one-step derivative into the three pieces that will later reappear in the score formula:

$$\partial_i \hat{\ell}_t = -\frac{1}{2} \partial_i \log \det S_t - \frac{1}{2} \partial_i (v_t^\top S_t^{-1} v_t) - \frac{1}{2} \partial_i (n_y \log 2\pi). \quad (15.3)$$

Since the last term is zero, the current-period score is controlled entirely by how S_t and v_t move. That is why the gradient path needs only two objects for the current likelihood contribution:

$$\dot{v}_t, \quad \dot{S}_t.$$

The cross-covariance derivative $\dot{C}_{xz,t}$ is not needed for $\hat{\ell}_t$. It is needed because $C_{xz,t}$ forms the gain K_t , and K_t updates m_t and P_t . Those updated sensitivities become the input to the next time step. Thus the derivative has two roles:

$$\text{score role: } (\dot{v}_t, \dot{S}_t) \mapsto \dot{\hat{\ell}}_t, \quad (15.4)$$

$$\text{propagation role: } (\dot{C}_{xz,t}, \dot{S}_t, \dot{v}_t) \mapsto (\dot{m}_t, \dot{P}_t). \quad (15.5)$$

Confusing these roles is a common implementation mistake. A program can appear to compute a correct one-step score while still propagating the wrong next-step derivative.

object	role	why it matters
$\dot{v}_t, \dot{S}_t$	current score	sufficient for the present innovation score in (15.24)
$\dot{C}_{xz,t}$	propagation only	needed because K_t depends on $C_{xz,t}$, even though the current score does not
$\dot{K}_t$	propagation only	transmits observation sensitivity into the posterior update
$\dot{m}_t, \dot{P}_t$	next-step state	become the differentiated inputs to the next prediction stage and the next factor derivative

There are four places where the derivative can stop:

1. P_{t-1} or P_t^- is not on the saved factor branch.
2. S_t is not positive definite.
3. A finite-difference perturbation changes the saved branch, grid, merge, or preprocessing policy.
4. A model derivative routine $D_x f_\theta$, $\partial_i f_\theta$, $D_x h_\theta$, $\partial_i h_\theta$, $\dot{Q}_\theta$, or $\dot{R}_\theta$ is missing or inconsistent with the value routine.

On those events the note does not patch the scalar by hidden repair. It returns a branch failure. This narrowness is exactly what lets the analytical gradient be checked by finite differences.

15.2 Square-Root Branch

This first stage differentiates the factor map itself. It is needed before any point sensitivity can be written down, because the placed cloud points depend on C_{t-1} and C_t^- . If this stage is wrong, every later derivative object is wrong even if the score algebra itself is correct. For the default unpivoted Cholesky branch, $P = LL^\top$ with positive diagonal and $C(P) = L$. The implementation convention is: always use the lower triangular Cholesky factor returned by the declared factor map, never a symmetric square root or pivoted alternative inside this scalar. Given symmetric $\dot{P}$, set

$$A = L^{-1} \dot{P} L^{-\top}. \quad (15.6)$$

Define the lower-triangular operator

$$\Phi(A)_{jk} = \begin{cases} A_{jk}, & j > k, \\ A_{jj}/2, & j = k, \\ 0, & j < k. \end{cases} \quad (15.7)$$

Then

$$\dot{L} = L\Phi(A). \quad (15.8)$$

Equation (15.8) is the governing implementation recipe for every factor derivative in this report. In particular, whenever the value path computes a covariance factor by $C_t = \text{chol}(P_t)$ or $C_t^- = \text{chol}(P_t^-)$, the gradient path computes the matching $\dot{C}_t$ or $\dot{C}_t^-$ by applying (15.6)–(15.8) to the corresponding $\dot{P}_t$ or $\dot{P}_t^-$. No alternative factor derivative is allowed inside the same scalar. Indeed, $\Phi(A) + \Phi(A)^\top = A$, so

$$\dot{L}L^\top + L\dot{L}^\top = L\{\Phi(A) + \Phi(A)^\top\}L^\top = \dot{P}. \quad (15.9)$$

If a different square-root map is used, the implementation must provide $\dot{C} = DC(P)[\dot{P}]$ for the same branch. That alternative is outside the scope of this report: every formula and algorithm below assumes the declared lower-triangular Cholesky branch $C(P) = \text{chol}(P)$.

15.3 Prediction Sensitivities

This stage moves the derivative from the previous carried Gaussian into the prediction cloud and the predictive moments. Its outputs, $\dot{m}_t^-$ and $\dot{P}_t^-$, are not yet enough for the current likelihood score, but they are necessary because both the next observation cloud and the next factor derivative depend on them. From (12.2),

$$\dot{\chi}_{t-1}^{(r)} = \dot{m}_{t-1} + \dot{C}_{t-1} \xi^{(r)}. \quad (15.10)$$

From (12.3),

$$\dot{a}_t^{(r)} = D_x f_\theta(\chi_{t-1}^{(r)}) \dot{\chi}_{t-1}^{(r)} + \partial_i f_\theta(\chi_{t-1}^{(r)}). \quad (15.11)$$

The predicted mean derivative is

$$\dot{m}_t^- = \sum_{r=1}^M w_r \dot{a}_t^{(r)}. \quad (15.12)$$

Let

$$d_t^{(r)} = a_t^{(r)} - m_t^-, \quad \dot{d}_t^{(r)} = \dot{a}_t^{(r)} - \dot{m}_t^-. \quad (15.13)$$

Then

$$\dot{P}_t^- = \dot{Q}_\theta + \sum_{r=1}^M w_r \left\{ \dot{d}_t^{(r)} d_t^{(r)\top} + d_t^{(r)} \dot{d}_t^{(r)\top} \right\}. \quad (15.14)$$

15.4 Observation Sensitivities

This stage converts predictive-state sensitivity into observation-side sensitivity. It is here that the derivative splits into the two objects needed for the current score, $\dot{v}_t$ and $\dot{S}_t$, and the extra object needed only for propagation, $\dot{C}_{xz,t}$. The logic is:

$$\dot{\chi}_t^{(r)} \mapsto \dot{z}_t^{(r)} \mapsto (\dot{z}_t, \dot{v}_t, \dot{\Delta}_t^{(r)}) \mapsto (\dot{S}_t, \dot{C}_{xz,t}). \quad (15.15)$$

Differentiate (12.6):

$$\dot{\chi}_t^{(r)} = \dot{m}_t^- + \dot{C}_t^- \xi^{(r)}. \quad (15.16)$$

Then

$$\dot{z}_t^{(r)} = D_x h_\theta(\chi_t^{(r)}) \dot{\chi}_t^{(r)} + \partial_i h_\theta(\chi_t^{(r)}). \quad (15.17)$$

The observation mean, residual, and centered residual derivatives are

$$\dot{\bar{z}}_t = \sum_{r=1}^M w_r \dot{z}_t^{(r)}, \quad (15.18)$$

$$\dot{v}_t = -\dot{\bar{z}}_t, \quad (15.19)$$

$$\dot{\Delta}_t^{(r)} = \dot{z}_t^{(r)} - \dot{\bar{z}}_t. \quad (15.20)$$

Differentiate S_t and $C_{xz,t}$:

$$\dot{S}_t = \dot{R}_\theta + \sum_{r=1}^M w_r \left\{ \dot{\Delta}_t^{(r)} \Delta_t^{(r)\top} + \Delta_t^{(r)} \dot{\Delta}_t^{(r)\top} \right\}, \quad (15.21)$$

$$\dot{C}_{xz,t} = \sum_{r=1}^M w_r \left\{ (\dot{\chi}_t^{(r)} - \dot{m}_t^-) \Delta_t^{(r)\top} + (\chi_t^{(r)} - m_t^-) \dot{\Delta}_t^{(r)\top} \right\}. \quad (15.22)$$

15.5 Innovation Score

At this point the current-period score problem is local. Once $\dot{v}_t$, $\dot{S}_t$, and the innovation solve variable are known, the pointwise cloud no longer appears explicitly in the one-step score. The proposition below is therefore the score-stage payoff of all earlier derivative work.

Proposition 2 (Fixed Gaussian innovation score). *Assume S_t is positive definite and the active branch is differentiable. Let u_t solve*

$$S_t u_t = v_t. \quad (15.23)$$

Then the derivative of (12.13) is

$$\dot{\ell}_t^{\text{FSGQ}} = -\frac{1}{2} \left[\text{tr}(S_t^{-1} \dot{S}_t) + 2\dot{v}_t^\top u_t - u_t^\top \dot{S}_t u_t \right]. \quad (15.24)$$

Proof. Use

$$d \log \det S = \text{tr}(S^{-1} dS) \quad (15.25)$$

and

$$dS^{-1} = -S^{-1}(dS)S^{-1}. \quad (15.26)$$

Therefore, with $u = S^{-1}v$,

$$d(v^\top S^{-1}v) = 2(dv)^\top u - u^\top (dS)u. \quad (15.27)$$

Substitute $dS = \dot{S}_t$ and $dv = \dot{v}_t$ into (5.3). $\square$

15.6 Posterior Sensitivity Propagation

If the note stopped at the innovation score, it would describe only how to differentiate one isolated likelihood contribution. The filter, however, continues in time. This stage therefore propagates the current observation-side sensitivity back into the carried Gaussian that seeds the next step. The gradient must also propagate $\dot{m}_t$ and $\dot{P}_t$ into the next time step. The implementation order is:

$$(\dot{C}_{xz,t}, \dot{S}_t, \dot{v}_t) \mapsto \dot{K}_t \mapsto (\dot{m}_t, \dot{P}_t) \mapsto \dot{C}_t \quad \text{at the next step through } DC(P_t)[\dot{P}_t]. \quad (15.28)$$

Differentiate $K_t = C_{xz,t} S_t^{-1}$:

$$\dot{K}_t = \dot{C}_{xz,t} S_t^{-1} - K_t \dot{S}_t S_t^{-1}. \quad (15.29)$$

Then

$$\dot{m}_t = \dot{m}_t^- + \dot{K}_t v_t + K_t \dot{v}_t, \quad (15.30)$$

$$\dot{P}_t = \dot{P}_t^- - \dot{K}_t S_t K_t^\top - K_t \dot{S}_t K_t^\top - K_t S_t \dot{K}_t^\top. \quad (15.31)$$

Equations (15.10)–(15.31) define the value-and-gradient recursion for the fixed scalar (14.2). At this point the reader should retain one structural summary: $\dot{v}_t$ and $\dot{S}_t$ close the current score, while $\dot{C}_{xz,t}, \dot{K}_t, \dot{m}_t, \dot{P}_t$ exist because the filter must carry its differentiated state forward in time.

16 One Boxed Mathematical Algorithm

FixedSGQF value and gradient for one fixed parameter component on one saved branch. The dot notation in this box means differentiation with respect to one declared parameter component θ_i . In particular, $\dot{m}_t, \dot{P}_t, \dot{C}_t^-, \dot{S}_t, \dot{C}_{xz,t}$, and $\dot{\ell}_t^{\text{FSGQ}}$ in this box are the one-component versions of the indexed quantities $\dot{m}_t^{(i)}, \dot{P}_t^{(i)}, \dot{C}_t^{-(i)}, \dot{S}_t^{(i)}, \dot{C}_{xz,t}^{(i)}$, and $\partial_i \dot{\ell}_t^{\text{FSGQ}}$ used in the full algorithm. This boxed routine is a compact summary of the full implementation-order algorithm in Section 18; when the two presentations differ in brevity, the longer end-to-end algorithm is the authoritative execution order.

1. Choose dimension $b = n_x$, level L , one-dimensional rules I_ℓ , merge tolerance, zero-weight threshold, and deterministic node ordering. Build and save the cloud $\mathcal{C} = \{(\xi^{(r)}, w_r)\}_{r=1}^M$ by (10.4)–(10.16), including duplicate merge, near-zero pruning, and final deterministic sorting. These cloud-construction choices are part of the saved branch record and therefore part of the declared scalar.
2. Save $y_{1:T}$, observation preprocessing, $m_0(\theta), P_0(\theta)$, the one-component sensitivity policy for θ_i , the full model objects $f_\theta, h_\theta, Q_\theta, R_\theta$, the declared lower-triangular Cholesky factor map $C(P) = \text{chol}(P)$, the symmetrize-then-veto rule, the eigenvalue thresholds (17.8), and model derivative routines. Together with the cloud-construction choices in Step 1, these objects form the saved branch record for the declared scalar.
3. Initialize $m_0, P_0, \dot{m}_0, \dot{P}_0$, the running scalar $\hat{\ell}_T^{\text{FSGQ}} \leftarrow 0$, and the running derivative $G_i \leftarrow 0$, where $G_i = \partial_i \hat{\ell}_T^{\text{FSGQ}}$ is the i -th component of the full score vector.
4. For $t = 1, \dots, T$, symmetrize P_{t-1} , then factor $P_{t-1} = C_{t-1} C_{t-1}^\top$ on the saved branch. For this fixed parameter component θ_i , compute $\dot{C}_{t-1} = DC(P_{t-1})[\dot{P}_{t-1}]$, and then run the prediction value equations (12.2)–(12.5) and prediction sensitivity equations (15.10)–(15.14).
5. If the symmetrized predictive covariance P_t^- fails the declared Cholesky or positive-definite test in (17.8), stop and report a predictive-branch failure.
6. Factor $P_t^- = C_t^- (C_t^-)^\top$ on the saved branch, compute $\dot{C}_t^- = DC(P_t^-)[\dot{P}_t^-]$, and then run the observation value equations (12.6)–(12.13), which produce the candidate one-step increment $\dot{\ell}_t^{\text{FSGQ}}$, together with the observation sensitivity equations (15.16)–(15.22).
7. If the symmetrized innovation covariance S_t fails the declared positive-definite branch test in (17.8), stop and report an innovation-branch failure.
8. Solve $S_t u_t = v_t$, so u_t is the innovation solve variable representing the action of S_t^{-1} on v_t without forming an explicit inverse. The observation sensitivity stage has already produced $\dot{z}_t$, hence $\dot{v}_t = -\dot{z}_t$, together with $\dot{S}_t$ on this same saved branch for this same parameter component θ_i . Compute the i -th one-step score component $\dot{\ell}_t^{\text{FSGQ}} = \partial_i \dot{\ell}_t^{\text{FSGQ}}$ by (15.24), update the running derivative by $G_i \leftarrow G_i + \dot{\ell}_t^{\text{FSGQ}}$, and after the innovation branch has passed update the running scalar by $\hat{\ell}_T^{\text{FSGQ}} \leftarrow \hat{\ell}_T^{\text{FSGQ}} + \dot{\ell}_t^{\text{FSGQ}}$.
9. Update $m_t, P_t, \dot{m}_t, \dot{P}_t$ using (12.14)–(12.16) and (15.29)–(15.31) for this fixed component θ_i . Then apply the same symmetrize-then-veto rule to the updated carried covariance: replace P_t by $(P_t + P_t^\top)/2$, and if the symmetrized matrix is not on the declared positive-definite factor branch, stop and report a carried-covariance failure.

17 Implementation Contract

object	definition	shape	reuse
$\xi^{(r)}$	standardized sparse-grid node	n_x	value and gradient
w_r	accumulated signed weight after duplicate merge	scalar	value and gradient
C_{t-1}, C_t^-	saved covariance factors on declared branch	$n_x \times n_x$	value, gradient, diagnostics
$\chi_{t-1}^{(r)}$	previous placed point	n_x	transition value and derivative
$a_t^{(r)}$	transition image	n_x	prediction moments
$\chi_t^{(r)}$	predictive placed point	n_x	observation value and derivative
$z_t^{(r)}$	observation image	n_y	observation moments
$\bar{z}_t, S_t, C_{xz,t}$	Gaussian projection moments	$n_y, n_y \times n_y, n_x \times n_y$	likelihood and update
v_t, K_t, u_t	innovation, Gaussian projection gain, and innovation solve variable with $S_t u_t = v_t$	$n_y, n_x \times n_y, n_y$	likelihood, update, gradient
$\dot{\chi}, \dot{a}, \dot{z}$	point sensitivities	matching objects	value gradient only
$\dot{m}, \dot{P}, \dot{S}, \dot{C}_{xz}$	moment and filter sensitivities	matching objects	value gradient and next step
data record	$y_{1:T}$, preprocessing, missing-data policy if any	metadata	value, gradient, finite difference
initial law	$m_0(\theta), P_0(\theta), \dot{m}_0, \dot{P}_0$	$n_x, n_x \times n_x$	value and gradient start
branch record	cloud, one-dimensional rules, merge tolerance, zero-weight rule, node ordering, preprocessing policy, initial-condition policy, declared Cholesky factor branch, symmetrize-then-veto rule, thresholds, accepted/failure stage-time pattern	metadata	same-scalar finite difference

17.1 Input, Output, Invariant, And Failure Contract

The full report-level routine has inputs

$$\begin{aligned}
\mathcal{I}_{\text{in}} = & (y_{1:T}, \theta, L, \{I_\ell\}_{\ell=1}^{L+b-1}, m_0(\theta), P_0(\theta), \\
& \dot{m}_0^{(1:p)}(\theta), \dot{P}_0^{(1:p)}(\theta), f_\theta, h_\theta, Q_\theta, R_\theta, \\
& D_x f_\theta, \partial_{1:p} f_\theta, D_x h_\theta, \partial_{1:p} h_\theta, \dot{Q}_\theta^{(1:p)}, \dot{R}_\theta^{(1:p)}, \\
& \text{merge tolerance, factor map, branch thresholds}).
\end{aligned} \tag{17.1}$$

It returns either

$$\left(\widehat{\ell}_T^{\text{FSGQ}}(\theta), \nabla_\theta \widehat{\ell}_T^{\text{FSGQ}}(\theta), \{m_t, P_t\}_{t=0}^T, \mathcal{D}_{\text{diag}} \right) \tag{17.2}$$

or a failure record

$$\mathcal{F} = (\text{time, stage, reason, diagnostic values}). \tag{17.3}$$

The invariants are:

$$\sum_{r=1}^M w_r = 1 \quad \text{up to construction tolerance,} \quad (17.4)$$

$$P_t^- = (P_t^-)^\top, \quad S_t = S_t^\top, \quad P_t = P_t^\top \quad \text{after explicit symmetrization,} \quad (17.5)$$

$$S_t \succ 0 \quad \text{on every accepted likelihood step,} \quad (17.6)$$

$$\mathcal{B}_{\text{value}} = \mathcal{B}_{\text{gradient}} \quad \text{same saved branch.} \quad (17.7)$$

The routine fails rather than repairs if

$$\lambda_{\min}(S_t) < \epsilon_S, \quad \lambda_{\min}(P_t^-) < \epsilon_P, \quad \lambda_{\min}(P_t) < \epsilon_P, \quad (17.8)$$

or if the declared factor map cannot be evaluated on the symmetrized covariance. An implementation may add a smoothed repair, but then the repair is a new scalar and needs a new derivative.

17.2 One-Step Value-Path Contract

At one accepted time step, the value path is the ordered map

$$\begin{aligned} (m_{t-1}, P_{t-1}, y_t, \theta, \mathcal{C}) &\mapsto C_{t-1} \mapsto \chi_{t-1}^{(r)} \mapsto a_t^{(r)} \mapsto (m_t^-, P_t^-) \\ &\mapsto C_t^- \mapsto \chi_t^{(r)} \mapsto z_t^{(r)} \mapsto (\bar{z}_t, S_t, C_{xz,t}) \\ &\mapsto (v_t, \hat{\ell}_t^{\text{FSGQ}}, K_t) \mapsto (m_t, P_t). \end{aligned} \quad (17.9)$$

The accepted map in (17.9) is conditioned on two successful branch checks: the predictive covariance P_t^- must pass the declared Cholesky and positive-definite test before observation-point placement, and the innovation covariance S_t must pass the declared branch test before the Gaussian projection update is accepted. The accepted outputs propagated to the next step are only (m_t, P_t) together with the global fixed branch record $\mathcal{B}$. Point clouds, moment arrays, gains, and diagnostic quantities may be cached for gradient reuse, but they are not part of the carried filtering state itself.

17.3 Default Numerical Constants

The constants below are defaults for diagnostics, not claims of universal validity:

quantity	default	role
duplicate merge tolerance	$10^{-12} \max(1, \ \xi\ _\infty)$	groups standardized nodes that should be identical under the declared rule
weight zero tolerance	10^{-14}	removes accumulated numerical-zero weights after merge
ϵ_S	10^{-10}	veto threshold for innovation covariance positive definiteness
ϵ_P	10^{-10}	veto threshold for carried and predictive covariance factors
ϵ_ℓ	$10^{-3} \max(1, \hat{\ell}_T)$	pilot level-ladder change threshold
finite-difference relative tolerance	10^{-5}	same-scalar gradient diagnostic
maximum branch-invalid FD rows	0 in final audit	branch changes mean the derivative is not being tested

These values are deliberately conservative. They are meant to reveal when the fixed branch is fragile. They are not tuning knobs for making a failing branch look smooth.

18 End-To-End Mathematical Algorithm

FixedSGQF value and gradient, with all branch exits.

1. Build the sparse-grid cloud from $b = n_x$, level L , and $\{I_\ell\}$. Use (10.4), (10.5), (10.12), (10.13), and (10.15). Save $\mathcal{C} = \{(\xi^{(r)}, w_r)\}_{r=1}^M$.
2. Check (17.4). If it fails, return $\mathcal{F} = (0, \text{cloud, weight-sum failure}, \dots)$.
3. Initialize $\widehat{\ell}_T^{\text{FSGQ}} \leftarrow 0$, $G_i \leftarrow 0$ for $i = 1, \dots, p$, and compute $m_0, P_0, \dot{m}_0^{(i)}, \dot{P}_0^{(i)}$.
4. For $t = 1, \dots, T$, symmetrize P_{t-1} . If the factor map $C(P_{t-1})$ fails, return a factor-branch failure. Otherwise set $C_{t-1} = C(P_{t-1})$ and for each parameter component compute $\dot{C}_{t-1}^{(i)} = DC(P_{t-1})[\dot{P}_{t-1}^{(i)}]$.
5. Place previous points by (12.2); compute their derivatives by (15.10) componentwise in $i = 1, \dots, p$.
6. Evaluate transition values (12.3) and derivatives (15.11).
7. Compute $m_t^-, P_t^-, \dot{m}_t^{-(i)}$, and $\dot{P}_t^{-(i)}$ by (12.4), (12.5), (15.12), and (15.14). Symmetrize P_t^- . If (17.8) fails for P_t^- , return a predictive covariance failure.
8. Factor P_t^- . Set $C_t^- = C(P_t^-)$ and compute $\dot{C}_t^{-(i)} = DC(P_t^-)[\dot{P}_t^{-(i)}]$ for each parameter component. Place observation points by (12.6); compute their derivatives by (15.16).
9. Evaluate observation values (12.7) and derivatives (15.17).
10. Compute $\bar{z}_t, \Delta_t^{(r)}, S_t, C_{xz,t}, v_t, \widehat{\ell}_t^{\text{FSGQ}}$ by (12.8)–(12.13). Symmetrize S_t . If S_t fails (17.8), return an innovation covariance failure.
11. Compute $\dot{\bar{z}}_t^{(i)}, \dot{v}_t^{(i)}, \dot{S}_t^{(i)}$, and $\dot{C}_{xz,t}^{(i)}$ by (15.18)–(15.22).
12. Solve $S_t u_t = v_t$. Add $\widehat{\ell}_t^{\text{FSGQ}}$ to $\widehat{\ell}_T^{\text{FSGQ}}$ only after the innovation branch has passed, and add the componentwise score to G_i only on that same accepted innovation branch. Reuse this same solved u_t for every parameter component i , and add the componentwise score from (15.24) with $\dot{v}_t = \dot{v}_t^{(i)}$ and $\dot{S}_t = \dot{S}_t^{(i)}$ to G_i .
13. Compute K_t first from (12.14). Then for each parameter component compute $\dot{K}_t^{(i)}$ from (15.29), and update $m_t, P_t, \dot{m}_t^{(i)}, \dot{P}_t^{(i)}$ by (12.15)–(12.16) and (15.30)–(15.31). Symmetrize P_t . If the carried covariance branch fails, return a carried covariance failure.
14. Return $(\widehat{\ell}_T^{\text{FSGQ}}, G_{1:p}, \{m_t, P_t\}, \mathcal{D}_{\text{diag}})$.

The algorithm is deliberately mathematical rather than Pythonic. Its purpose is to say what a Python implementation must compute, save, and refuse to change if it wants the reported gradient to be the derivative of the reported value.

18.1 Formula Inventory For The Value Path

For implementation review, the value path can be located in the report as follows.

value-path item	report location
symbol and dimension table	Section 6.1
exact filtering target	Section 6
Gaussian projection moment equations	Section 7
standardized sparse-grid cloud construction	Section 10
merge convention and stored cloud semantics	Section 10.4 and Section 11
per-step value inputs and outputs	Sections 12, 17.1, and 17.2
prediction equations	(12.2)–(12.5)
observation and update equations	(12.6)–(12.16)
one-step log-likelihood increment	(12.13)
implementation-order value/gradient algorithm	Section 18
branch checks and numerical conventions	Sections 6.1, 17.1, and 17.3

19 Finite-Difference Same-Scalar Check

The finite-difference check must call the same value routine with the same saved branch $\mathcal{B}$. For parameter component i , use

$$D_i(h) = \frac{\widehat{\ell}_T^{\text{FSGQ}}(\theta + he_i; \mathcal{B}) - \widehat{\ell}_T^{\text{FSGQ}}(\theta - he_i; \mathcal{B})}{2h}. \quad (19.1)$$

Use the step ladder

$$h \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\} \max\{1, |\theta_i|\}. \quad (19.2)$$

For every h , the $+h$ and $-h$ evaluations must reuse the same saved structural branch record $\mathfrak{B}$: the same cloud, one-dimensional rules, preprocessing policy, initial-condition policy, declared Cholesky factor branch, merge tolerance, zero-weight rule, node ordering, and symmetrize-then-veto rule. The numerical initial values $m_0(\theta \pm he_i)$, $P_0(\theta \pm he_i)$ are recomputed by that same policy. Each evaluation also returns its realized accepted/failure stage-time pattern. A row is branch-valid only if both perturbations match the base evaluation in that full branch-identity record and in the realized accepted/failure stage-time pattern through every likelihood contribution included in the scalar comparison. Equal factor family but different first failure location counts as branch mismatch. Equal first failure location but different accepted-prefix pattern also counts as branch mismatch. If either side hits a different branch or returns a veto, that row is marked branch-invalid and is not interpreted as a derivative check. For valid rows, compare $D_i(h)$ against the analytical score G_i by

$$e_i(h) = \frac{|D_i(h) - G_i|}{\max\{1, |D_i(h)|, |G_i|\}}. \quad (19.3)$$

The diagnostic passes when at least two consecutive valid rows have $e_i(h) \leq 10^{-5}$ and the smaller-step row is no worse than ten times the larger-step row. This is a numerical diagnostic, not a theorem: failure means one of three things is likely. The derivative formula is wrong, the value and gradient paths are not using the same scalar, or the branch is too nonsmooth or ill-conditioned at that point.

Implementation-facing FD pseudocode summary. A routine of the form `same_scalar_fd_check(i, hladder,  $\theta$ , ...)` should:

1. evaluate the value path at $\theta \pm h e_i$ using the same saved structural branch metadata,
2. mark a row branch-invalid if either side changes the structural branch or the realized accepted/failure stage-time pattern,
3. compute the central difference only for branch-valid rows,
4. compare that central difference against the analytical score with (19.3), and
5. report whether two consecutive valid rows satisfy the declared tolerance rule.

This summary is the operational finite-difference contract for an implementation; the equations above remain the authoritative mathematical definition.

A scalar numerical trace. Let $X \sim \mathcal{N}(0, 1)$, $h_\theta(x) = \theta x$, $R = 1$, and $y = 2$. Any fixed Gaussian rule with exact second moment gives

$$\bar{z} = 0, \quad S(\theta) = 1 + \theta^2, \quad \hat{\ell}(\theta) = -\frac{1}{2} \left\{ \log(1 + \theta^2) + \frac{4}{1 + \theta^2} + \log(2\pi) \right\}. \quad (19.4)$$

At $\theta = 1$,

$$G = -\frac{1}{2} \left\{ \frac{2}{2} - \frac{4 \cdot 2}{2^2} \right\} = 0.5. \quad (19.5)$$

Central differences give:

h	$D(h)$	G	$ D(h) - G $
10^{-1}	0.5008108250	0.5	8.10825×10^{-4}
10^{-2}	0.5000083311	0.5	8.33108×10^{-6}
10^{-3}	0.5000000833	0.5	8.33331×10^{-8}
10^{-4}	0.5000000008	0.5	8.33722×10^{-10}

A cloud-sensitive nonlinear trace. The previous trace tests the innovation-score algebra. To exercise the fixed cloud, use the two-dimensional level-2 cloud in Section 11 and the nonlinear observation

$$h_\theta(x_1, x_2) = \theta(x_1^2 + x_2^2), \quad X \sim \mathcal{N}(0, I_2), \quad R = 1, \quad y = 3. \quad (19.6)$$

For the merged cloud in Section 11,

$$\sum_r w_r(x_{1,r}^2 + x_{2,r}^2) = 2, \quad \sum_r w_r(x_{1,r}^2 + x_{2,r}^2)^2 = 6. \quad (19.7)$$

Therefore

$$\bar{z}(\theta) = 2\theta, \quad S(\theta) = 1 + 2\theta^2, \quad v(\theta) = 3 - 2\theta, \quad (19.8)$$

if the parameter scales the whole nonlinear observation linearly. To exercise the parameter through placed nonlinear values more sharply, set instead

$$h_\theta(x_1, x_2) = \theta^2(x_1^2 + x_2^2), \quad (19.9)$$

which gives

$$\bar{z}(\theta) = 2\theta^2, \quad S(\theta) = 1 + 2\theta^4, \quad v(\theta) = 3 - 2\theta^2. \quad (19.10)$$

At $\theta = 1$, the score from (15.24) is

$$G = -\frac{1}{2} \left[\frac{8}{3} + 2(-4)\frac{1}{3} - \frac{1}{3^2}8 \right] = \frac{4}{9}. \quad (19.11)$$

Central differences for the saved cloud scalar give:

h	$D(h)$	G	$ D(h) - G $
10^{-1}	0.5200303321	0.4444444444	7.55859×10^{-2}
10^{-2}	0.4452146703	0.4444444444	7.70226×10^{-4}
10^{-3}	0.4444521481	0.4444444444	7.70369×10^{-6}
10^{-4}	0.4444445215	0.4444444444	7.70368×10^{-8}

This example depends on the merged sparse-grid fourth moment in (19.7). A duplicate-merge bug or a different saved cloud changes $S(\theta)$, G , and the finite-difference table.

20 Diagnostics, Accuracy, Memory, And Performance Tests

Fixed SGQF should be tested as an approximate deterministic likelihood, not as an exact posterior method. The minimum validation suite is:

1. **Construction checks.** Verify duplicate-node accumulation, weight totals, signed-weight counts, and deterministic cloud reproduction for small (b, L) cases such as Section 11.
2. **Linear-Gaussian recovery.** For affine f_θ, h_θ , the Gaussian projection recursion should match the Kalman filter up to quadrature and floating-point tolerance.
3. **Quadratic scalar cell.** Use (5.5) to show both the moment accuracy and the posterior-shape limitation.
4. **Same-scalar finite differences.** Run (19.1) for representative parameters and step sizes.
5. **Signed-weight stability.** Record the minimum eigenvalues of symmetrized P_t^- , S_t , and P_t , plus any branch veto.
6. **Memory and runtime scaling.** Record $M_{b,L}$, model evaluations, wall time, and peak memory as b, L, T change.
7. **Accuracy ladder.** Compare fixed SGQF against tensor-product GHQ in small dimension, Kalman truth in linear-Gaussian models, and particle or TT diagnostics in nonlinear models where those references are available.

Passing these checks supports the statement that the implementation evaluates and differentiates the declared approximate scalar. It does not prove exact posterior accuracy.

20.1 Mathematical Test Models

The validation suite should use models that expose different failure modes. The following models are mathematical specifications for future code tests. They are ordered to mirror the report’s argumentative burden: first recover exact or near-exact reference behavior where possible, then expose the Gaussian-surrogate limitation, then test the cloud-specific and high-dimensional claims that later feed into the method-selection argument of Section 22.

Model A: linear-Gaussian recovery. Let

$$X_t = A_\theta X_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_\theta), \quad (20.1)$$

$$Y_t = H_\theta X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_\theta). \quad (20.2)$$

The Kalman filter gives the exact likelihood and exact filtering moments. A correct FixedSGQF implementation should match the Kalman mean, covariance, and innovation scalar up to floating-point tolerance because all required moments are at most quadratic in the Gaussian coordinate. This test checks moment assembly, covariance factors, likelihood algebra, and derivative propagation, but it does not stress nonlinear quadrature accuracy.

Model B: scalar quadratic observation. Use

$$X_t = \rho X_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q), \quad (20.3)$$

$$Y_t = X_t^2 + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, r). \quad (20.4)$$

This model is small enough to compare against dense numerical integration. For small r and positive observations, the exact posterior can have two lobes. The test should report both moment agreement and posterior-shape failure. It is the guardrail against accidentally describing FixedSGQF as an exact nonlinear posterior filter.

Model C: cloud-sensitive two-dimensional nonlinear observation. Let

$$X_t = \rho X_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, qI_2), \quad (20.5)$$

$$Y_t = \theta^2(X_{t,1}^2 + X_{t,2}^2) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, r). \quad (20.6)$$

The two-dimensional level-2 cloud in Section 11 gives the explicit moments in (19.7). This test catches duplicate-node and signed-weight errors because the fourth moment changes when the cloud is assembled incorrectly.

Model D: block-local high-dimensional nonlinear model. For $b = md$, partition $x \in \mathbb{R}^b$ into m blocks $x_{[j]} \in \mathbb{R}^d$. Let

$$X_t = A_\theta X_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_\theta), \quad (20.7)$$

$$Y_{t,j} = \alpha_j^\top x_{t,[j]} + \beta_j \|x_{t,[j]}\|^2 + \gamma_j x_{t,[j]}^\top B_j x_{t,[j+1]} + \varepsilon_{t,j}, \quad (20.8)$$

with $j = 1, \dots, m$ and $x_{[m+1]} = x_{[1]}$. This model has controlled low-order interactions. It is designed to test whether moderate sparse-grid levels exploit interaction sparsity and whether point count, memory, and runtime grow as expected. Diagnostics should report $b, L, M_{b,L}, T$, model evaluations per step, peak memory, wall time, and branch failures.

Model E: deliberately hard all-coordinate interaction. Let

$$Y_t = \delta \prod_{j=1}^b (1 + \alpha X_{t,j}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, r). \quad (20.9)$$

This model is not expected to be easy for a moderate sparse grid. It probes the failure mode where high-order coordinate interactions matter. A good validation report should not hide this case; it should show when the level ladder or covariance diagnostics reject the fixed grid.

Model F: Jia–Xin–Cheng orbit-style benchmark. Jia, Xin, and Cheng evaluate SGQF on an orbit estimation problem [1, Section 4]. This note does not reproduce their full experiment, but a future implementation should include an orbit-style benchmark because it is the source paper’s main empirical setting. The test should record the state dimension, observation geometry, noise covariances, chosen sparse-grid level, point count, RMSE, likelihood diagnostics, and finite-difference branch status. This benchmark supports source-faithful comparison; it is not a theorem of general performance.

20.2 Large-Scale Report Template

For each large-scale run, record

$$\begin{aligned} \mathcal{R} = & (b, L, M_{b,L}, T, \text{number of parameters, random seed,} \\ & \text{model evaluations per step, peak memory, wall time,} \\ & \min_t \lambda_{\min}(P_t^-), \min_t \lambda_{\min}(S_t), \min_t \lambda_{\min}(P_t), \\ & \widehat{\ell}_T, \|\nabla \widehat{\ell}_T\|, \text{finite-difference table, branch failures}). \end{aligned} \quad (20.10)$$

Accuracy claims should be made only against a named comparator:

$$\text{error} = \begin{cases} |\widehat{\ell}_T - \ell_T^{\text{Kalman}}|, & \text{linear-Gaussian test,} \\ |\widehat{\ell}_T - \ell_T^{\text{dense}}|, & \text{small dense quadrature test,} \\ \text{RMSE or posterior diagnostic against PF/TT reference,} & \text{nonlinear large test.} \end{cases} \quad (20.11)$$

Memory and performance do not rescue a failed accuracy or branch diagnostic. They are interpreted after the veto checks. These validation objects therefore set the stage for the final two interpretive tasks of the report: first, to say what adaptive sparse-grid selection can still contribute once same-scalar requirements are imposed, and second, to decide how FixedSGQF should be judged against neighboring algorithmic lanes.

21 Adaptive Sparse Grids As Grid Design

Singh et al. [2] choose sparse-grid indices adaptively by using admissible index sets, local error indicators, active and old index sets, a global error estimate, and a tolerance. That is useful for designing a cloud before inference. For HMC, however, the live target cannot silently change its grid as θ moves. The safe interpretation is

$$\text{adaptive pilot chooses a cloud} \implies \text{freeze the cloud} \implies \text{run FixedSGQF}. \quad (21.1)$$

If the adaptive index set is rerun inside the likelihood, the value path is a piecewise-defined algorithm with discrete changes. The formulas in Section 15 do not differentiate those changes. This is why adaptive sparse-grid selection enters the note as an upstream design aid rather than as the selected differentiable likelihood lane. With that restriction now explicit, the report can finally compare FixedSGQF against neighboring methods on the exact criteria that matter for selection.

22 Relation To Neighboring High-Dimensional Filtering Proposals

The previous sections have fixed the mathematical object this note is trying to carry: one Gaussian surrogate (m_t, P_t) together with the deterministic same-scalar likelihood

$$\hat{\ell}_T^{\text{FSGQ}}(\theta) = \sum_{t=1}^T \hat{\ell}_t^{\text{FSGQ}}(\theta). \quad (22.1)$$

This section is therefore not a general survey. It asks a narrower mathematical question: among nearby approximate filtering lanes, which constructions define a usable deterministic surrogate scalar, which preserve an analytical derivative of that same scalar on a fixed branch, and which avoid the dense-growth burden that would make the lane unusable in the intended high-dimensional regime?

Quick comparison table.

method	carried object	moment engine	same-scalar derivative lane	main limitation
EKF	one Gaussian	local linearization	yes, for the linearized scalar	weak nonlinear moment fidelity
UKF/CKF	one Gaussian	fixed low-order deterministic points	yes	lower-order deterministic rule
Dense GHQF	one Gaussian	dense tensor-product Gaussian quadrature	yes	exponential point growth
FixedSGQF	one Gaussian	fixed sparse-grid quadrature	yes	still one Gaussian; signed-weight branch sensitivity
Adaptive sparse grid	one Gaussian	parameter-dependent active grid	only if frozen first	target changes when the live grid changes
Particle / TT lanes	richer non-Gaussian object	samples or density representation	different target class	heavier approximation object, different contract

For any deterministic Gaussian-surrogate lane M , write its induced scalar as

$$\ell_T^{(M)}(\theta) = \sum_{t=1}^T \log \mathcal{N}(y_t; \bar{z}_t^{(M)}(\theta), S_t^{(M)}(\theta)). \quad (22.2)$$

and its moment engine as the operator

$$(\bar{z}_t^{(M)}, S_t^{(M)}, C_{xz,t}^{(M)}) = \mathcal{Q}_M(f_\theta, h_\theta, m_{t-1}, P_{t-1}, Q_\theta, R_\theta). \quad (22.3)$$

The comparison is then mathematical rather than rhetorical: nearby Gaussian filters differ by the operator $\mathcal{Q}_M$ that feeds the same Gaussian update algebra, while particle and tensor-density methods differ already at the level of the carried object.

For a deterministic point rule with nodes $\xi_r^{(M)}$ and weights $w_r^{(M)}$, that operator has the explicit form

$$\bar{z}_t^{(M)} \approx \sum_{r=1}^{M_M} w_r^{(M)} h_\theta(m_t^- + C_t^- \xi_r^{(M)}), \quad (22.4)$$

$$S_t^{(M)} \approx R_\theta + \sum_{r=1}^{M_M} w_r^{(M)} \Delta_{t,r}^{(M)} \Delta_{t,r}^{(M)\top}, \quad \Delta_{t,r}^{(M)} := h_\theta(m_t^- + C_t^- \xi_r^{(M)}) - \bar{z}_t^{(M)}, \quad (22.5)$$

$$C_{xz,t}^{(M)} \approx C_t^- \left(\sum_{r=1}^{M_M} w_r^{(M)} \xi_r^{(M)} \Delta_{t,r}^{(M)\top} \right). \quad (22.6)$$

Thus the decisive question is not whether these methods are all “Gaussian,” but what kind of moment engine they use and what that engine implies for the scalar and its derivative.

What disqualifies a lane for this note? A lane is not selected here if any one of the following fails:

- (i) a declared deterministic approximate scalar objective,
- (ii) an analytical derivative of that same scalar on a fixed smooth branch, (22.7)
- (iii) a representation that avoids dense tensor-product growth in the intended regime.

This is not a theorem of universal method superiority. It is the local choice architecture of this report. Conditions (i)–(iii) are used here as hard vetoes: a method that fails one of them is not selected for this lane. Once those conditions are met, the later comparisons concern which eligible lane best realizes the report’s strong preferences.

EKF and local linearization. The extended Kalman filter carries the same kind of Gaussian object (m_t, P_t) , but its moment engine is not (22.4)–(22.6). Instead it substitutes local linear surrogates,

$$f_\theta(x) \approx f_\theta(m) + F_\theta(x - m), \quad h_\theta(x) \approx h_\theta(m^-) + H_\theta(x - m^-), \quad (22.8)$$

so that the predictive and observation moments are inherited from Jacobian closure rather than from explicit nonlinear point evaluation. In particular, the moment operator becomes

$$\mathcal{Q}_{\text{EKF}} \approx (h_\theta(m_t^-), H_\theta P_t^- H_\theta^\top + R_\theta, P_t^- H_\theta^\top), \quad (22.9)$$

which is computationally cheap but sacrifices the explicit nonlinear pointwise moment information preserved by (22.4)–(22.6). The present report does not reject EKF as incoherent; it rejects it because the selected lane insists on evaluating the nonlinear maps themselves under a declared cloud.

Low-order sigma-point and cubature lanes. UKF and CKF remain viable deterministic Gaussian alternatives because they also construct a scalar of the form (22.2). Their difference lies in the point design,

$$\mathcal{Q}_{\text{low}} \sim \sum_{r=1}^{M_{\text{low}}} \tilde{w}_r F(\tilde{\xi}_r), \quad M_{\text{low}} = O(n_x), \quad (22.10)$$

where the low point count is attractive but the deterministic rule remains a lower-order moment design. FixedSGQF instead chooses the sparse-grid rule

$$\mathcal{Q}_{\text{SGQF}} \sim \sum_{r=1}^{M_{b,L}} w_r F(\xi^{(r)}), \quad M_{b,L} = O(b^{L-1}) \quad \text{for fixed } L, \quad (22.11)$$

The mathematical tradeoff is therefore visible in two dimensions at once:

$$M_{\text{low}} = O(n_x) \quad \text{versus} \quad M_{b,L} = O(b^{L-1}), \quad (22.12)$$

while the corresponding moment operators satisfy

$$\mathcal{Q}_{\text{low}} \neq \mathcal{Q}_{\text{SGQF}} \quad \text{except in special low-order cases where the same moments are matched.} \quad (22.13)$$

So the comparison is not “deterministic versus nondeterministic,” but low-order low-point deterministic moment closure versus higher-structure sparse-grid deterministic moment closure. The selected lane is willing to pay more than $O(n_x)$ points in exchange for a richer deterministic moment structure while still avoiding the tensor-product explosion of dense GHQF. The decisive point for this note is not that low-order sigma-point rules are invalid, but that once the project asks for a deterministic Gaussian-surrogate lane with a visibly recorded cloud, merge, and branch identity, the sparse-grid construction carries more of the moment-engine burden in the object itself rather than in a smaller rule whose adequacy must be argued indirectly.

Standard and tensor-product GHQF. Dense Gaussian quadrature remains the cleanest low-dimensional reference, but it fails the present lane in the intended regime because the point count grows as

$$M_{\text{tensor}}(b, L) = s_L^b, \quad (22.14)$$

where s_L is the one-dimensional point count at the chosen level. By contrast, the sparse-grid cloud in this report has

$$M_{\text{SGQF}}(b, L) = M_{b,L} = O(b^{L-1}) \quad \text{for fixed } L. \quad (22.15)$$

The note does not claim that sparse-grid quadrature is uniformly more accurate than dense GHQF. It claims something narrower: if the lane is defined by deterministic Gaussian-surrogate likelihood evaluation in a high-dimensional regime, then dense GHQF is too expensive to be the selected lane and should remain a comparator or small-dimension reference.

Live adaptive sparse-grid selection. Adaptive sparse-grid methods are still useful, but not as the selected same-scalar derivative lane. If the active index set changes with θ , then the scalar itself becomes

$$\ell_T^{\text{adap}}(\theta) = \sum_{t=1}^T \ell_t(\theta, \mathcal{I}_t(\theta)), \quad (22.16)$$

where $\mathcal{I}_t(\theta)$ is the parameter-dependent active index set. Its formal derivative is then only local to regions where the index set is constant:

$$\partial_i \ell_T^{\text{adap}}(\theta) = \sum_{t=1}^T \partial_i \ell_t(\theta, \mathcal{I}_t(\theta)) \quad \text{only on regions where } \mathcal{I}_t(\theta) \text{ is locally constant.} \quad (22.17)$$

That is a different target class from the fixed scalar of this note. The role of adaptation here is therefore upstream cloud design, not the selected downstream same-scalar likelihood lane.

Particle and tensor-density lanes. Particle and TT methods differ already at the level of the carried object. Particle filtering carries weighted samples; squared TT carries a richer non-Gaussian density approximation. In schematic form,

$$\text{particle lane: carried object } \{x_t^{(r)}, \omega_t^{(r)}\}_{r=1}^N, \quad \text{TT lane: carried object } \hat{p}_t^{\text{TT}}(x), \quad (22.18)$$

whereas FixedSGQF carries only (m_t, P_t) together with the surrogate scalar. These methods are not ruled out because they are weak. They are retained as complementary lanes because they preserve posterior structure that one Gaussian surrogate cannot preserve. The present note is therefore not selecting the method with the broadest posterior-shape fidelity; it is selecting the deterministic Gaussian-surrogate lane with the cleanest same-scalar likelihood-and-gradient contract.

Mathematical selection summary. The comparison therefore closes not with a slogan but with the following local selection statement:

$$\text{Selected lane} = \text{deterministic Gaussian-surrogate scalar} + \text{fixed-branch analytical derivative} + \text{sparse-grid moment equations} \quad (22.19)$$

This equation is not a theorem of universal optimality. It is the summary of the local choice architecture built in this report. FixedSGQF is selected because, among nearby deterministic Gaussian candidates, it is the lane that jointly preserves: a declared deterministic scalar, an analytical derivative of that same scalar on a fixed branch, an explicit approximation boundary, and a cost story less explosive than dense GHQF while more stable than live adaptive-grid selection. That is the precise sense in which this note chooses FixedSGQF.

23 Where Each Construction Comes From

construction	support	use in this note
state-space and conceptual filtering equations	Jia et al. [1, Section 2]	Sections 6–7
Gaussian approximation filter moment structure	Jia et al. [1, Eq. 15–23] plus Proposition 1	moment projection and update equations
one-dimensional and tensor-product Gaussian quadrature	Jia et al. [1, Section 2.2.1]	Section 9
sparse-grid formula and index band	Jia et al. [1, Eq. 26–29]	Section 10
point/weight generation and duplicate accumulation	Jia et al. [1, Algorithm 1]	fixed cloud $\mathcal{C}$ and toy grid
quadrature exactness and UKF relation	Jia et al. [1, Theorems 3.1–3.2]	source-scope orientation, not posterior guarantee
adaptive sparse-grid mechanics	Singh et al. [2, Section 3]	offline grid-design interpretation
fixed scalar and analytical gradient comparison with squared TT	BayesFilter project derivation Zhao–Cui companion note and Zhao and Cui [3]	Sections 14–15 Section 22

24 Notation And Formula Inventory

For implementation review, the note can be navigated by the following inventory.

item	report location
scientific problem and lane choice	Sections 2–3
exact object versus approximate object	(3.2), Section 5
global notation and dimensions	Section 6.1
Gaussian projection equations	Section 7
source-order sparse-grid construction	Section 10
merge convention and stored cloud semantics	Section 10.4, Section 11
value-path recursion	Section 12, Section 17.2, Section 18.1
fixed scalar and saved-branch contract	Section 14, (14.2), (14.4)
factor derivative convention	Section 15, subsection ‘Square-Root Branch’, equations (15.6)–(15.8)
gradient dependency chain and roles	(15.1), (15.4), (15.5)
prediction and observation sensitivities	(15.10)–(15.22)
innovation-score derivative	Proposition 2, (15.24)
posterior sensitivity propagation	(15.28), (15.29)–(15.31)
one-component summary algorithm	Section 16
full implementation-order algorithm	Section 18
worked numeric oracle	Section 13
finite-difference same-scalar diagnostics	Section 19
neighboring-method positioning	Section 22
validation models and report template	Section 20, Section 20.2

25 Conclusion

This report has addressed a specific scientific problem: high-dimensional nonlinear filtering in a regime where exact posterior propagation is not a practical computational object, but where a deterministic and analytically differentiable surrogate may still be scientifically valuable. The proposal advanced here is FixedSGQF, a lane that carries one Gaussian surrogate, approximates its required nonlinear moments by a fixed sparse-grid cloud, and differentiates the resulting deterministic same-scalar likelihood on a declared branch.

What the report has constructed is therefore precise. It has reconstructed the Gaussian-projection sparse-grid filtering value path in source order, made the saved branch explicit, given a branch-conditioned analytical gradient of the same scalar that the value routine evaluates, provided a compact worked numeric oracle, and written the value and gradient recursions in implementation order. A reader should now be able to identify the exact carried state, the exact accumulated scalar, the exact derivative contract, and the exact places where a branch veto occurs.

The report has also been explicit about what it does *not* claim. It does not claim exact nonlinear likelihood evaluation. It does not claim that sparse-grid exactness for selected Gaussian moments implies exact non-Gaussian posterior shape. It does not claim that one carried Gaussian can preserve multimodality, heavy asymmetry, or all-coordinate interaction structure whenever those features matter scientifically. It does not replace richer non-Gaussian density-approximation lanes such as Zhao–Cui squared TT.

The strongest scientific case for FixedSGQF is therefore not generic novelty but a specific

combination of properties. The cloud is declared, the covariance-factor convention is declared, the branch-veto logic is declared, the scalar is deterministic on that branch, and the analytical gradient differentiates that same declared scalar. Those properties are exactly the criteria the report used to distinguish FixedSGQF from its nearest deterministic Gaussian alternatives: more explicit nonlinear point evaluation than EKF, a more transparent sparse-grid cloud contract than UKF/CKF, a less explosive cost story than dense GHQF, and a cleaner same-scalar derivative contract than live adaptive grid selection. This makes the method well suited to settings where reproducibility, inspectability, and gradient-based parameter learning matter, and where the filtering task can tolerate a Gaussian-surrogate carried object. In regimes with lower effective interaction order, the sparse-grid moment engine can also be scientifically plausible without requiring a full tensor-product explosion.

The main caution is equally clear. FixedSGQF should not be oversold as a full non-Gaussian posterior method. When posterior shape itself is central — for example under strong multimodality, hard global interactions, or branch fragility — the deterministic Gaussian-surrogate lane becomes scientifically too narrow, and richer sample-based or density-based methods deserve priority. That is not a defect in the present report; it is the explicit boundary that keeps the proposal honest.

A fair review-panel judgment should therefore be the following. FixedSGQF is a coherent, mathematically explicit, self-contained, and implementation-ready high-dimensional filtering lane. It is narrower than a full non-Gaussian density approximation, but that narrowness buys a transparent deterministic surrogate and a branch-conditioned analytical gradient that can be audited, implemented, and checked directly. The approval claim remains bounded: this report is not asserting that FixedSGQF is the best nonlinear filter overall, but that it is the selected deterministic Gaussian-surrogate lane for the explicit approximate likelihood-and-gradient objective developed here. On that basis, this report is sufficiently thorough and persuasive to justify approval of FixedSGQF as one serious lane in a broader high-dimensional filtering program.

References

- [1] Bin Jia, Ming Xin, and Yang Cheng. Sparse-grid quadrature nonlinear filtering. *Automatica*, 48(2):327–341, 2012. doi: 10.1016/j.automatica.2011.08.057.
- [2] Abhinoy Kumar Singh, Rahul Radhakrishnan, Shovan Bhaumik, and Paresh Date. Adaptive sparse-grid gauss-hermite filter. *arXiv preprint arXiv:1803.09272*, 2018.
- [3] Yiran Zhao and Tiangang Cui. Tensor-train methods for sequential state and parameter learning in state-space models. *Journal of Machine Learning Research*, 25:1–51, 2024.